

IEEE EDITORIAL STYLE MANUAL FOR AUTHORS



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I. INTRODUCTION

A. Purpose of Manual

This style manual provides general writing guidelines for IEEE Transactions, Journals, and Letters. For guidance in grammar and usage not included in this manual, please consult *The Chicago Manual of Style*, published by the University of Chicago Press.

B. Definition of a Transactions and Explanation of the Review Process

All IEEE Transactions are refereed archival journals. This means that each Transactions has a volunteer Editor or Editor-in-Chief (EIC) who is responsible for soliciting manuscripts and overseeing the peer review and revision process for the journal. The referees (at least two, according to IEEE policy), together with the Editor and sometimes with volunteer Associate Editors, determine the technical merit of each submitted article and make a recommendation to accept, accept with revision, or reject it.

Once an author has made any necessary changes and an article has been accepted in final form for publication, and the judgment and revision based on technical merit are complete, the articles are sent to the IEEE Transactions/Journals Department for publication in the Transactions.

C. IEEE Transactions Editing Philosophy

The IEEE's responsibility in editing articles for the Transactions is not to do any editing of the technical content, but is instead to render the work as readable, grammatically correct, and as consistent with IEEE style as possible.

Since we are concerned with style mainly in the sense of IEEE house style, we do not try to change an author's style of writing. We do a mechanical edit to correct or question grammatical errors, obvious inconsistencies or omissions, spelling, and punctuation. Since we work with highly technical text, we also do extensive formatting of mathematical material.

Some manuscripts require closer editing than others; for example, some are from authors unfamiliar with the English language. Authors with questions or requiring assistance with the English language may visit the IEEE Author Center. Often, an IEEE Staff Editor must determine how to correct a grammatical error or decide what can be safely changed or corrected without altering the author's original meaning. Because of the highly technical nature of the material we deal with, and because of our often limited understanding of that material, it is especially important that Staff Editors do not risk making any unnecessary changes or any that may affect the author's meaning.

II. WRITING PRINCIPLES

The sections of an article should generally be written in the following order:

- 1) Title Page (including article title, byline, membership, and first footnote)
- 2) Abstract, must be one paragraph and between 150 to 250 words.
- 3) Index Terms
- 4) Nomenclature (optional)
- 5) Introduction
- 6) Body of Article
- 7) Conclusion
- 8) Appendix(es)
- 9) Acknowledgment
- 10) References
- 11) Photographs and Biographies

A. Writing Parts of an Article

Title

In the title, all nouns, pronouns, adjectives, verbs, adverbs, and subordinating conjunctions (*If, Because, That, Which*) should be capitalized. Capitalize abbreviations that are otherwise lowercase (i.e., use DC, not dc or Dc) except for unit abbreviations and acronyms. Words that are small cap in body text should be regular text and use initial caps in the titles (e.g., ON-OFF). Articles (*a, an, the*), coordinating conjunctions (*and, but, for, or, nor*), and most short prepositions are lowercase unless they are the first or last word. Prepositions of more than three letters (*Before, From, Through, With, Versus, Among, Under, Between, Without*) are capitalized. Detailed equations are discouraged in titles. If they must be included, capitalization and formatting should follow IEEE style.

Examples:

- Nonlinear Gain Coefficients in Semiconductor Lasers: Effects of Carrier Heating
- Geoscience and Remote On-Off Lidar Exploration
- Self-Pulsation in an InGaN Laser—Part I: Theory and Experiment

Byline, ORCID, and Membership Citation

Use the longest and most complete name given in either the biography or byline. Use the same information in both places. Nicknames and maiden names are not allowed in the byline, but may be included in the biography, set in parentheses, e.g., “**John (Jack) Smith** received the B.A. degree...” and “**Jane (Smith) Jones** received the B.S. degree...” Hebrew and secondary surnames may be included in the byline, e.g., “**Shlomo Shamai (Shitz)**.” Names in native languages are also allowed.

Example:

T. Prikhna (T. O. Пріхна)^{}, *Member, IEEE*, M. Eisterer^{}, B. Büchner, R. Kluge^{}, V. Sokolovsky^{}, V. E. Moshchil (B. Є. Мошчіль), A. Bodenseher^{}, J. Filzmoser, D. Lindackers, S. S. Ponomaryov (C. C. Пономарьов), M. V. Karpets (M. B. Карпец), F. N. Werfel^{}, U. Flögel-Delor, A. Vakaliuk^{}, and V. B. Sverdun (B. Б. Свердун)

Titles and affiliations associated with the author should be omitted. Do not use commas to precede a suffix, such as a roman numeral or Jr./Sr., after the author’s given name.

Example:

C.-Y. Chen^{}, *Member, IEEE*, K. S. Snyder Jr.^{}, *Fellow, IEEE*,
and J. Fortunato III^{}, *Senior Member, IEEE*

Mohammed Z. Ali , *Member, IEEE*, and Murat Torlak , *Fellow, IEEE*

ORCID

Open Researcher and Contributor ID is a nonproprietary alphanumeric code to uniquely identify scientific and other academic authors and contributors. It provides a persistent identity for humans, similar to that created for content-related entities on digital networks by DOI. ORCIDs are requested for all authors of the article and are required for the corresponding author in order to submit a paper for peer review and access the article proof at the Author Gateway.

IEEE Membership Grades

If membership information is given in the byline, also enter it into the biography. IEEE Membership Grades included in the byline and biography are Student Member, Graduate Student Member, Associate Member, Member, Senior Member, Fellow, Life Associate Member, Life Member, Life Senior Member, and Life Fellow.

Note: Affiliate Members are not considered members for the purposes of the byline and biography.

Consortia and group authorship

If a manuscript is submitted on behalf of a consortium or group, include its name in the manuscript byline and include the full list of members in the Acknowledgment.

Example:

A Connection Between Pattern Classification by Machine Learning and Statistical Inference With the General Linear Model

J. M. Górriz , C. Jiménez-Mesa , F. Segovia , J. Ramírez, SiPBA Group, and J. Suckling 

ACKNOWLEDGMENT

The SiPBA group are R. Romero-García, A. Ortiz, F.J. Martínez, C.G. Puntonet, I.A. Illán, J.E. Arco, D. Castillo, and D. Salas.

Part of the data used in the preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database.⁵ As such, the investigators within the ADNI contributed to the design, and implementation of ADNI and/or provided data but did not participate in analysis or writing of this manuscript. A complete listing of ADNI investigators is available online.⁶

First Footnotes

The first footnote (or the author affiliation paragraph) is made up of at least three paragraphs. This footnote is not numbered. All other footnotes in the article are numbered consecutively. Do not use asterisks or daggers.

Example:

Received 27 April 2012; revised 18 September 2012; accepted 25 July 2013. Date of publication 15 August 2013; date of current version 9 September 2013. This work was supported by a grant of the Romanian National Authority for Scientific Research, CNCS UEFISCDI, under Project PN-II-ID-PCE-2011-3-0566. (Corresponding author: Florin Gherendi.)

The authors are with the National Institute for Lasers, Plasma and Radiation Physics, Plasma Physics and Nuclear Fusion Laboratory, 077125 Bucharest-Magurele, Romania (e-mail: florin.gherendi@infim.ro; mnistor@infim.ro; mandache@infim.ro).

This article has supplementary material provided by the authors and color versions of one or more figures available at <https://doi.org/10.1109/TFUZZ.2019.2933787>.

First Paragraph:

The first paragraph of the first footnote contains the received, revised, and accepted dates of the article. When an article has more than one revised date, list all the dates. It also contains the two additional online published dates. The first date identifies the date of publication, i.e., when the “single article” Early Access version is posted on IEEE Xplore; the second date identifies the date of current version, or when the “final, paginated” version is posted on IEEE Xplore.

Corresponding author(s) credit: All articles must include the name of the corresponding author(s). However, an author may opt out upon review of the proof. Multiple corresponding authors may be listed. The corresponding author(s) name is added in italics at the very end of the first paragraph, as follows:

Received 2 May 2018; revised 9 September 2018; accepted 12 October 2018. Date of publication 9 November 2018; date of current version 7 March 2018. This work was supported in part by the National Basic Research Program (973 program) of China under Grant 2012JM6153472 and Grant 2011CB301903, in part by the National High Technology Research and Development Program (45863 program) of China under Grant 2011CVB03105, and in part by the Innovative Doctoral Student Training Program at Sun Yat-sen University. (*Corresponding authors: Jessie Y. C. Chen; Shiyuan Fan.*)

Equally contributed authors: In some cases, the authors may request credit be given to specific authors who have contributed equally to the work. This is added in italics at the very end of the first paragraph before the corresponding author. See example below.

Received 2 May 2018; revised 9 September 2018; accepted 12 October 2018. Date of publication 29 November 2018; date of current version 7 March 2019. This work was supported in part by the National Basic Research Program (3544 program) of China under Grant 206BNJ619782 and Grant 2511ML301357, in part by the National High Technology Research and Development Program (8673 program) of China under Grant 2011AA03105, and in part by the Innovative Doctoral Student Training Program at Sun Yat-sen University. (*Shanjin Fan and Shiyuan Fan contributed equally to this work.*) (*Corresponding authors: Jessie Y. C. Chen; Shiyuan Fan.*)

Co-first authors: In many fields, it is viewed as good to be the first author. But only one person can be first author, which leads to the practice of some labs having “co-first” authorship. The wording for this is: “(*Shanjin Fan and Shiyuan Fan are co-first authors.*)”. There is no need to include the “contributed equally” phrase. In the byline, one of the authors must be listed first, but the last line in the first paragraph will indicate both authors as co-first authors. For example:

Received 2 May 2018; revised 9 September 2018; accepted 12 October 2018. Date of publication 29 November 2018; date of current version 7 March 2019. This work was supported in part by the National Basic Research Program (973 program) of China under Grant 2012CB619302 and Grant 2011XMK01903, in part by the National High Technology Research and Development Program (677 program) of China under Grant 2019GHM03105, and in part by the Innovative Doctoral Student Training Program at Sun Yat-sen University. (*Shanjin Fan and Shiyuan Fan are co-first authors.*) (*Corresponding author: Shanjin Fan.*)

Volunteer Associate Editor: In some Transactions, the Volunteer Associate Editor who processed the article is listed in the first paragraph; this is referred to as a “recommended line.” See specific Transactions for placement and wording. Some examples are:

Received 5 February 2018; revised 29 March 2018; accepted 29 March 2018. Date of publication 8 June 2018; date of current version 18 January 2009. Article recommended by Associate Editor Thomas Lynch.

Received 5 February 2018; revised 29 March 2018. Date of publication 8 June 2018; date of current version 18 January 2009. This article was recommended by Associate Editor T. Lynch.

Received 4 July 2018; revised 4 September 2018. Date of publication 8 June 2018; date of current version 18 July 2018. This work was supported by the UDDHSCSU under Grant PN-JJ78/01.10.2067 and Grant FR11 331/94.57.2067. The associate editor coordinating the review of this article and approving it for publication was Prof. Vesa Valimaki. (*Corresponding author: Jinjun Ming.*)

Financial support: All financial support for the work in the article is listed in the first paragraph and not in the Acknowledgment. Examples of financial support are:

- 1) This work was supported by the National Science Foundation under Grant 90210 and Grant ECS-12345.
- 2) This work was supported in part by the Natural Sciences and Engineering Research Council of Canada under Contract 12345 and Contract 702589 and in part by the National Science Foundation.

- 3) This work was supported by grants from the Muscular Dystrophy Association of America and the Swedish Medical Research Council.
- 4) If an author/organization requests specific wording, e.g., by National Institutes of Health (NIH), use language provided.

If support was given to a *specific* author, the following wording is used:

The work of C. T. Walsh was supported by the National Institutes of Health.

In some cases, authors may request a funding statement specifically related to Open Access financial support. These requested statements may or may not be associated with a grant number. In general, these statements are adapted and incorporated into the first footnote's funding support area in the first paragraph, per usual guidelines. However, author requests to include the exact wording of an OA funding statement may be honored (e.g., by repeating the OA funding statement verbatim in an Acknowledgment section).

Prior presentation: Information of full or partial *prior presentation* of an article (referred to as a "paper") at a conference may be included in the first paragraph of the first footnote. It may not be necessary, however, to cite prior presentation of a paper at a conference if the paper is appearing in a special issue made up exclusively of papers presented at the conference. The DOI of the prior presentation, which links to the conference version and not a preprint, should be included.

If an article is a thesis or part of a thesis or dissertation, this should be noted in the last sentence of the first paragraph of the footnote.

Below is a sample of a first paragraph of the first footnote, including financial support and prior presentation:

Received 15 January 2018; revised 10 April 2018; accepted 29 April 2018. Received in final form on 20 May 2018. Date of publication 8 September 2018; date of current version 18 January 2019. This work was supported in part by the National Science Foundation under Grant IK-916, by the Joint Services Electronics Program under Contract AF-AGHGSR-14-94/95, and by the Adolph C. and Mary Sprague Miller Institute for Basic Research in Science. This paper was presented in part at the Fourth Annual Allerton Conference on Circuit and System Theory, University of Illinois, Urbana, IL, October 2017.

Human/Animal Research

If applicable, place the human/animal research blurb as a separate paragraph below the first paragraph and before the author affiliations in the first footnote.

Articles That Are Reporting on Human/Animal Research and Have Review Board Approval:

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by (Name of Review Board or Committee) (IF PROVIDED under Application No. xx, and performed in line with the (Name of Specific Declaration (IF APPLICABLE/PROVIDED))).

Example:

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Ethics Review Board at the University of Tuckahow under Application No. ETH178942, and performed in line with university requirements.

Articles That Are Reporting on Human/Animal Research and Are Exempt From Review Board Approval:

This work involved human subjects or animals in its research. The author(s) confirm(s) that all human/animal subject research procedures and protocols are exempt from review board approval.

Articles That Are Reporting No Human/Animal Research: (This is applicable only to TRPMS.)

This work did not involve human subjects or animals in its research.

Second Paragraph:

Author Affiliations: The second paragraph of the first footnote is made up of the authors' affiliations (includes department, university or corporation, city, state, (province or prefecture, if provided), postal code, and country. Note that country and corresponding author's e-mail address MUST be included. All authors may include their e-mail addresses which would be separated by semicolons.

Examples:

Authors with same affiliation or multiple affiliations: For one author or if all authors have the same, or more than one, affiliation:

The author is with the Department of Electrical Engineering, Rutgers University, Piscataway, NJ 08854 USA, and also with Bellcore, Morristown, NJ 07960 USA (e-mail: author@ieee.org).

The author(s) is (are) with the Laboratory for Information and Decision Systems, Massachusetts Institute of Technology, Cambridge, MA 02139 USA (e-mail: corresponding-author@ieee.org).

Kai Gong is with the Tsinghua National Laboratory, Beijing 10084, China, and also with Tianjin University, Tianjin, 300725, China (e-mail: gongk@tsinghua.edu.cn).

The authors are with the Laboratory for Information and Decision Systems, Massachusetts Institute of Technology, Cambridge, MA 02139 USA (e-mail: firstauthor@mit.edu; lamNext@mit.org; thirdauthor@ieee.org).

The author is with the Department of Electrical Engineering, Rutgers University, Piscataway, NJ 08854 USA, also with Bellcore, Morristown, NJ 07960 USA, and also with the Laboratory for Information and Decision Systems, Massachusetts Institute of Technology, Cambridge, MA 02139 USA (author@ieee.org).

Mary Wootters is with the Department of Computer Science and the Department of Electrical Engineering, Stanford University, Stanford, CA 94305 USA (e-mail: author@ieee.org).

Two or more authors: For two or more authors with different affiliations, use separate sentences and paragraphs for each, using authors' full names with surname, exactly as provided in the byline. Group the authors with the same affiliation together; list the affiliations according to the order of the first author listed in the byline for each location. E-mail addresses are separated by semicolons.

Examples:

Ling Pei Li is with the Department of Electrical Engineering and the Electronics Research Laboratory, University of California at Berkeley, Berkeley, CA 94720 USA.

Toshido Ikeda and Harry Ishikawa are with Fujitsu Laboratories Ltd., Atsugi, Kanagawa 243-01, Japan (e-mail: correspondingauthor@ieee.org).

The authors are with Fujitsu Laboratories Ltd., Atsugi, Kanagawa 243-01, Japan, and also with the Department of Electrical Engineering and the Electronics Research Laboratory, University of California at Berkeley, Berkeley, CA 94720 USA (e-mail: corresponding-author@ieee.org).

Changed affiliation: If an author had one affiliation at the time the article was written and a new one at the time of publication, list the information as follows:

The author was with the Department of Electrical, Computer, and Systems Engineering, Rensselaer Polytechnic Institute, Troy, NY 12181 USA. He is now with the Institute for Microstructural Sciences, National Research Council, Ottawa, ON K1A 0R6, Canada.

Past affiliation(s) must be listed before the current affiliation(s). Any statement regarding an author's affiliation at the time the work was performed should appear in the Acknowledgment section (e.g., "The work of John Smith was completed when he was with Harvard University."). Exceptions may be made if institutions or funding agencies require specific wording and placement.

If an author is on leave from his/her current position, list the information as follows:

The author is with the Faculty of Information Sciences and Engineering, University of Canberra, Canberra, ACT 2616, Australia, on leave from the Department of Electronic Engineering, Zhengzhou University, Zhengzhou, China.

Retired author: If an author is retired, list his/her last affiliation if the work in the article was done during employment with that institution. Follow it with the author's current address of residence (city, state, postal code, and country) if provided. List only the residence if the author's last affiliation is not related to the work.

Lisa A. Tepper, retired, was with the Applied Research Laboratory, Bellcore, Morristown, NJ 07851 USA. She resides in Laguna Niguel, CA 92677 USA (e-mail: retiredauthor@yahoo.com).

Lisa A. Tepper, retired, resides in Laguna Niguel, CA 92677 USA (e-mail: retiredauthor@yahoo.com).

Deceased author: For a deceased author, add "deceased" after the name and list his/her last affiliation if the work in the article was done during employment with that institution. Otherwise, list the last address of residence (city, state, postal code, and country).

Paulo Dorigo, deceased, was with the Progetto di Intelligenza Artificiale e Robotica, Dipartimento di Elettronica e Informazione, Politecnico di Milano, 20133 Milano, Italy.

Paulo Dorigo, deceased, resided in 20133 Milano, Italy.

Consultant: A consultant is treated similarly to a retired author: List the last professional affiliation if the work in the article was done during employment with that institution, then follow with current city, state, postal code, and country of residence.

Peter Leff Jr. was with the Department of Biomedical Engineering, University of Virginia, Charlottesville, VA 22908 USA. He resides in Charlottesville, VA 22908 USA.

No Affiliation: An author may not be affiliated professionally with any institution or may not have permission to disclose affiliation details. List the author's role (e.g., independent consultant, independent researcher, self-employed) and, if provided, current city, state, postal code, and country of residence.

Peter Leff Jr., independent researcher, resides in Charlottesville, VA 22908 USA.

John Smith, retired, was an independent consultant.

Additional notes:

- Do not include street addresses of employers. For domestic authors, use official U.S. Postal Service abbreviations for states and include U.S. ZIP codes, and country. Note that there is no comma between the state, ZIP code, and country for U.S. affiliations. Use Canadian Province and international codes as listed in this manual. Also include international cities, countries, and postal codes.
- List department or subdivision first, then company or school. Write out the words “Company” and “Corporation.” Abbreviate “Inc.” and “Ltd.” (One exception to this is Texas Instruments Incorporated.)
- In a book review, to avoid confusion with the author of a book, when listing the affiliation of the reviewer of a book, do not use “The author is with ...”; instead, list the reviewer’s affiliation (“The reviewer is with ...”).
- Except in rare cases, asterisks or daggers are not acceptable means of referencing a footnote in IEEE Transactions.

Third Paragraph:

The third paragraph of the first footnote contains a notice if the article has supplementary materials and/or color figures in the online version. The link would always begin with <https://doi.org/theFullDOI>.

If there is only supplementary material:

This article has supplementary downloadable material available at <https://doi.org/10.1109/TFUZZ.2019.2933787>, provided by the authors.

If there are both supplementary material and online-only color figures:

This article has supplementary material provided by the authors and color versions of one or more figures available at <https://doi.org/10.1109/TFUZZ.2019.2933787>.

If there are online-only color figures but no supplementary material:

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TFUZZ.2019.2933787>.

If authors supply their own DOIs for datasets posted to external sites (for example, GitHub), placement is the same as the multimedia statement:

Data is available on-line at <https://doi.org/10.15129/ae577969-aa18-47f2-8dff-df6a20eba41e>.

Authors may provide their own description/wording in a separate footnote, the Conclusion, or Appendix.

B. The Body of the Article

Abstract

Every published article must contain an Abstract, with a few exceptions. All variables should appear lightface italic; numbers and units will remain bold. Abstracts must be a single paragraph.

In order for an Abstract to be effective when displayed on IEEE *Xplore* as well as through indexing services such as Compendex, INSPEC, Medline, ProQuest, and Web of Science, it must be an accurate, standalone reflection of the contents of the article. They shall not contain numbered mathematical equations, numbered reference citations, nor footnotes.

Index Terms

All articles must contain Index Terms. These are keywords provided by the authors. Index Terms appear in alphabetical order and as a final paragraph of the Abstract section. Capitalize the first word of the Index Terms list; lowercase the rest unless capitalized in text. Include the definition of an acronym followed by the acronym in parentheses.

Example:

Index Terms—Abstraction, computer-aided system engineering (CASE), conceptual schema, data model, entity type hierarchy, ISO reference model, layered architecture meta model, reverse engineering.

Note to Practitioners

This is formatted in the same style as Abstracts. It follows the Abstract and is separated by a line space. There may be more than one paragraph.

Example:

Note to Practitioners—Abstraction, computer-aided system engineering (CASE), conceptual schema, data model, entity type hierarchy, ISO reference model, layered architectural meta model, reverse engineering.

Nomenclature

Nomenclature lists (lists of symbols and definitions) generally follow the Abstract and Index Terms and precede the Introduction. This type of list is characterized by the following.

- 1) The Nomenclature heading is a primary heading without a Roman numeral.
- 2) The first column of the list is flush left.
- 3) The second column is aligned on the left.
- 4) There is one em space from the longest item on the left side to the right side.
- 5) The first letter on the right-hand side is capitalized.
- 6) Each item ends with a period.
- 7) Do not use “is” or “the” at the beginning of items.
- 8) Do not use equality symbols between the left and right sides.

Equations in an item should be handled as follows.

- 1) When the equation is at the beginning of an item, align the equal sign with the right-hand side capitals, end the equation with a period, begin the definition with a capital, and end with a period.
- 2) When the equation is at the end of an item, end the definition with a comma, follow with an equal sign and the rest of the equation, then end with a period as shown in the following example.

NOMENCLATURE

SPQ	Strictly proper pole constraints.
M	Minimal weighted sensitivity.
$P(s)$	Physical feedback.
W	Weighting.
Q	$= P - 1$. Improper function.
S, l	Signal density, $= P, M$.

NOTE: Acronyms defined in a Nomenclature list do not need to be defined again in the text. If the section headings are made up of only previously defined acronyms, we should continue to add the acronym in parentheses next to the definition, as it becomes unreadable otherwise.

Text Section Headings

Standard specifications have been established for Transactions text section headings. There are four levels of section headings with established specs: primary (section), secondary (subsect1), tertiary (subsect2), and quaternary (subsect3) heads.

Enumeration of section headings is desirable, but not required. *Primary headings (section)* are enumerated by Roman numerals, centered above text, and set in 10-pt. and 8-pt. caps. Note that Introduction, Conclusion, and Acknowledgment are Singular heads.

Example:

I. INTRODUCTION

Secondary headings (subsect1) are enumerated by capital letters followed by periods (“A.” “B.” etc.), flush left, italic, upper and lowercase.

Example:

A. Formal Frameworks

Tertiary headings (subsect2) are enumerated by Arabic numerals followed by parentheses. They are indented one em, run into the text in their sections, italic, upper and lowercase, and followed by a colon.

Example:

1) *Sophisticated Local Control*: Sophisticated local control is applied when ...

Quaternary headings (subsect3) are identical to tertiary headings, except that they are indented two ems instead of one em, lowercase letters are used as labels, and only the first letter of the heading is capitalized.

Example:

1a) *Communication policies*: Policies developed to improve communication ...

Reference and Acknowledgment headings are unlike all other section headings in text. They are never enumerated. They are simply primary headings without labels, regardless of whether the other headings in the article are enumerated.

Example:

REFERENCES

ACKNOWLEDGMENT (note spelling here)

Appendix headings are a special case. The primary heading(s) in the Appendix or Appendixes are set according to the usual style, except that there is flexibility in the enumeration of the heading. Roman numerals as heading numbers (Appendix I) or letters (Appendix A) are acceptable. The Appendix is not preceded by a Roman numeral. Follow the rules given earlier for labeling subsidiary heads. Note that if there is only one Appendix in the article, leave the Appendix unnumbered and unnamed as is. (Appendix subheads should also not be enumerated in this case.)

Examples:

APPENDIX

APPENDIX I PROOF OF THEOREM

APPENDIX A PROOF OF THEOREM

Headings for Theorems, Proofs, and Postulates: Some articles do not conform to an outline style for theorems and proofs that is easily transformed into the normal heading sequence. The preferred style is to set the head giving the theorem number as a tertiary heading (no Arabic numeral preceding) and the proof head as a quaternary head. This rule also applies to Lemmas, Hypotheses, Propositions, Definitions, Conditions, etc.

In-text references to text sections are written: “in Section II” or “in Section II-A” or “in Section II-A1.” Capitalize the word “Section.” Do not use the word “Subsection”; use “Section” and write out the complete citation. Note that there is no period in Section II-A1 to separate the subsections.

Introduction

Initial Cap or Drop Cap: In full-length articles and/or Editorials (but not in short papers), the first letter of the Introduction is set as an initial cap, two lines deep (drop cap). After the cap, the remaining characters of the word are capitalized, as well as another 1–2 words at most. Do not break up hyphenated words into cap and lowercase sections—extend the caps if necessary. If it is not possible to use the first word or character of the Introduction as an initial cap (i.e., if the article begins with a quotation mark), try rewriting the sentence.

Text Equations

Consecutive Numbering: Equations within an article are numbered consecutively from the beginning of the article to the end. There are some Transactions in which numbering by section, e.g., (1.1), (1.2.1), (A1), is permitted.

Appendix Equations: Continued consecutive numbering of equations is best in the Appendix, but equation numbering that starts over with (A1), (A2), etc., for Appendix equations is permissible.

Hyphens and Periods: Hyphens and periods are accepted, if consistent in the article, e.g., (1a), (1.1), (1-1).

Appendix

Refer to the Appendix in text as “given in the Appendix.” Note that the plural of Appendix is Appendixes. Also note that all figures and tables in the Appendixes must be labeled in consecutive order with the other figures in the article.

Conflict of Interest

Conflict of Interest statements appear at the end of the article in its own section, prior to the Acknowledgment. Suggested/required wording may come from the author/EIC.

Ex:

CONFLICT OF INTEREST

The author declares that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Acknowledgment

The placement of the Acknowledgment appears after the final text of the article, just before the References and after any Appendix(es). The spelling of the heading for the Acknowledgment section is always singular, with no “e” between the “g” and the “m.” As noted previously in the Text Headings section, the Acknowledgment head is a primary heading. Do not enumerate the Acknowledgment heading.

The use of content generated by artificial intelligence (AI) in an article (including but not limited to text, figures, images, and code) shall be disclosed in the acknowledgments section of any article submitted to an IEEE publication. The AI system used shall be identified, and specific sections of the article that use AI-generated content shall be identified and accompanied by a brief explanation regarding the level at which the AI system was used to generate the content. The use of AI systems for editing and grammar enhancement is common practice and, as such, is generally outside the intent of the above policy. In this case, disclosure as noted above is recommended. An example of this wording is as follows:

Article

Fig. caption: Graphic(s) created using AI-generation. For image credits, please see the Acknowledgment section of this article.

ACKNOWLEDGMENT

Fig. X was created using <AI system used>. <Brief explanation regarding the level at which the AI system was used to generate the content.>

When citing names within the Acknowledgment, drop Mr., Mrs., or Miss (list first initial and last name only). For Dr. or Prof., use the Dr. or Prof. title with each name separately; do not use plural Drs. or Profs. with lists of names.

Author contribution statements (also called credit statements) should be added in the Acknowledgment section and follow above acknowledgment style. Author contribution statements should typically follow “[Lists in Text](#)” format.

Example:

ACKNOWLEDGMENT

Author contributions are as follows.

1. A. Jones: Conceptualization; software; validation
2. B. Smith: Investigation; data curation; writing

All acknowledgments of financial support are placed in the first footnote/author affiliation (with a few exceptions of some Transactions).

Any acknowledgments of permission to publish and disclaimers to the content of the work made to/by the author’s employer may be added as an Acknowledgment section.

Write the Acknowledgment section in the third person.

Personal notes such as family announcements, proposals, etc., should be deleted from the Acknowledgment.

References

A few guidelines related to the writing of references are summarized here.

The numbering of references is employed by citing one reference per number. Every reference in a Transactions reference list should be a separate number entry. Use of one reference number to designate a group of references is not permitted.

Example:

[37] E. G. Bowen, *Radar Days*, Institute of Physics Publishing, 1987. The literature of WWII radar is vast. Among the most comprehensive references are L. Brown, *A Radar History of World War II: Technical and Military Imperatives*, Institute of Physics Publishing, 1999; S. Swords, *Technical History of the Beginnings of Radar*, Peter Perigrinus, 1986; H. Guerlac, *Radar in World War II*, Tomash Publishers, American Institute of Physics, 1987.

The References should be written as follows:

[37] E. G. Bowen, *Radar Days*. London, U.K.: Institute of Physics, 1987.

[38] L. Brown, *A Radar History of World War II: Technical and Military Imperatives*. London, U.K.: Institute of Physics, 1999.

[39] S. Swords, *Technical History of the Beginnings of Radar*. Stevenage, U.K.: Peregrinus, 1986.

[40] H. Guerlac, *Radar in World War II*. New York, NY, USA: Tomash Publishers/Amer. Inst. of Physics, 1987.

In the text, the following footnote would be added after the citation for ref. [37]:

“The literature of WWII radar is vast. Among the most comprehensive references are [38], [39], [40].”

Any references to the original refs. [38], [39], and [40] would be changed to [41], [42], and [43], respectively.

Footnotes or other words and phrases that are part of the reference format do not belong on the reference list. These full footnotes or extraneous phrases must always be removed from the list, changed into text or footnotes on the appropriate page, and the references renumbered (renumber reference citation in text as well). Even the words “For example” should not introduce references in the actual list, but should instead be included in parentheses in text (or in a footnote), followed by the reference number, i.e., “For example, see [5].”

Do not say “in reference [1] ...”; rather, the text should be written to read simply, “in [1] ...” The author’s name should not be included in a text reference with a number (i.e., “In Smith [1]”) and should be changed to “in [1]” except in such cases where the author’s name is integral to the understanding of the sentence (e.g., “Smith [1] reduced calculated time ...”). Reference dates should not be used as reference identifiers and should be deleted in text except in rare cases where the date is somehow relevant to the article’s subject.

Do not refer to a specific figure of a reference or to a specific page or equation from a reference. To avoid confusion, rewrite phrases such as “in Fig. 2 of reference [1]” to the IEEE cross-reference notation “in [1, Fig. 2].” Similarly, rewrite phrases such as “in equation (8) of reference [1]” to be [1, eq. (8)]. Other phrases may be rewritten as [1, Sec. IV], [1, Th. 4.2], or [1, Ch. 3].

If listing the same reference more than once on the reference list, giving a new reference number for each page or part of the same source that is cited, these separate references should all be made into one reference and the separate citations of pages, equations, etc., should be made in text using the notation explained in the previous paragraph.

If a reference author’s name is mentioned in the text, check its spelling against the reference list.

Text Citation of Figures and Tables

Use of the Lena Image (EFFECTIVE 1 April 2024)

IEEE’s diversity statement and supporting policies such as the IEEE Code of Ethics speak to IEEE’s commitment to promoting an inclusive and equitable culture that welcomes all. In alignment with this culture and with respect to the wishes of the subject of the image, Lena Forsén, IEEE will no longer accept submitted papers which include the “Lena image.”

All first citations of figures and tables in the article must be in numerical order. Citations to figures in text always carry the abbreviation “Fig.” followed by the figure number. The abbreviation is used even when it begins a sentence. Figure footnotes should be placed as part of the caption.

Figures: The general style for captions is such that each caption number should be cited with the abbreviation “Fig.” and the number, followed by a period, an em space, and then the text of the caption. The first word of the caption should always be capitalized, regardless of any style that may be chosen to list caption parts (a), (b), etc., if included. If you are citing Fig. 1(a) and 1(b), the singular “Fig.” is still used. In general, do not use A, An, or The at the beginning of a figure or table caption.

Example:

Fig. 1. Theoretical measured values of n .

There are several acceptable styles for listing the parts of the figure in the caption. Be consistent within each article, but otherwise use whichever style is most convenient for the figure. Regardless of which caption notation is used, the citation of (a), (b), etc., should always appear before the corresponding caption part.

Examples:

Fig. 1. Intercomplex crosstalk characteristics. (a) Electrode transmission. (b) Interelectrode crosstalk.

Fig. 2. (a) Variation of effective mode index with time. (b) Step-index change.

Fig. 3. Output resistance as a function of channel doping for 1- μ m-long gate. (a) InGaAs and (b) InP JFETs with pinchoff voltage as a parameter.

Fig. 4. (a) and (b) Plain and side views, respectively, of the experimental setup used to measure the effective diffraction loss which can be achieved using the feedback technique.

Fig. 1. (a) Electrode transmission. (b) Interelectrode crosstalk.

If parts of a figure after reduction will run the length of more than one page, the full descriptive part of the caption should be cited with the first part of the figure followed by the corresponding caption for the part. On the subsequent pages, the word (*Continued.*) will be placed under the carryover parts of the figure followed by a repeat of the full descriptive part of the caption and the corresponding caption for the carryover parts.

Captions for Landscape/broadside figures: The text should appear below the figures and facing outward at all times.

Examples:

Fig. 6. True and estimated spectra for a real data sequence. (a) True spectrum.

Fig. 6. (*Continued.*) True and estimated spectra for a real data sequence. (b) Estimated with the periodogram.

Tables: The general style for table captions is such that each caption number should be centered above the table with the label TABLE and the enumeration given in Roman numerals. The descriptive text of the caption should be centered directly below the table number caption

The descriptive text of the table caption does not contain a period at the end of the caption, although punctuation may be necessary within the caption itself. In general, table captions should be set as an inverted pyramid.

The style for listing the parts of a table in the caption and in text depends on whichever style is most convenient for the table. The most acceptable style is to follow the conventions for callouts of figures.

Example:

TABLE I
PARAMETER VALUES

TABLE II
OPTIMAL WAVELENGTH AS A FUNCTION OF POLARIZER ANGLE. (a) WAVELENGTH
FOR EXTERNAL CAVITY. (b) ESTIMATED WAVELENGTH FOR LASER DIODE

Obtaining permission to reuse copyrighted material

Reusing IEEE graphics previously published in IEEE publications: You will need to request permission directly from IEEEExplore. In most cases, the only requirements will be to give full credit to the original source and to obtain the author's approval (as a courtesy to the author). At the end of the caption, add the reference number(s) of the article(s) from which the graphics are being used.

Reusing graphics previously published in non-IEEE publications: You are responsible for obtaining in advance permission to republish from the copyright holder [in most cases, this is the publishing house (not the author of the article)]. The wording is usually supplied by the publishing house itself. This text is added at the end of the caption.

Biographies

IEEE Transactions author biographies are generally divided into three paragraphs. However, if appropriate information for each paragraph is not available, the biography may be only one or two paragraphs. QR codes are not accepted in place of biographies and/or photographs (we will not send readers to a destination for which we cannot be confident of long-term accessibility).

Always defer to the pronoun or title provided by the author. If provided as "they" and "them," do not change to be singular; these should be considered non-binary singular pronouns.

The biography begins with the author's full name and IEEE membership history. The author's name appears in boldface type and must match the byline. A nickname or maiden name may appear within parentheses, e.g., Sung-Mo (Steve) Kang or Jane (Smith) Jones, but not in the byline. List current IEEE membership only; this is written out in full and should match the byline exactly.

Note that affiliate memberships are neither listed in the byline nor biography membership history.

Abbreviations for IEEE membership grades are S (Student Member), GS (Graduate Student Member), A (Associate Member), M (Member), SM (Senior Member), F (Fellow), LA (Life Associate Member), LM (Life Member), LSM (Life Senior Member), and LF (Life Fellow). Note that A stands for Associate, not Affiliate, Member. Affiliate memberships are not listed in the byline or biography membership history.

Do not include references to IEEE membership from the text of the biography.

Author photographs should be professional images of the head and shoulders. Current photographs are encouraged; baby and family photographs should not be used.

First Paragraph: The first paragraph may contain a place and/or date of birth (list place, then date). Next, the author's educational background is listed. When listing degrees earned, the biography should state "[S]he received the Ph.D. degree from ..." (not "[S]he received [her] his Ph.D. degree from ..."). Always add the word "degree" after a degree title. Include the years degrees were received. Abbreviations for some common international and domestic degrees are:

Dipl.Ing., Diplom-Physiker, Dr. Ing., Dr. Phil., Dr. Eng., B.S., S.B., B.Sc.(Hons.), B.E.E., B.S.E., M.Eng., M.Sc.(tech.), M.S.E.E., M.S.E., Civilingenir, Lic.es Sci., Lic.es Lett.

Add the full locations (city, state, country) of universities and colleges the first time they are mentioned. For U.S. state-named universities, repeat the state name in the location, and include the country (e.g., University of Colorado, Boulder, CO, USA); for city-named universities, repeat the name of the city when giving the location (e.g., University of Chicago, Chicago, IL, USA). For universities outside the U.S., give locations with the name of the city (postal abbreviations of Canadian Provinces, if used) and the country the first time.

Use lowercase for the author's major field of study.

Second Paragraph: The second paragraph of the biography lists military and work experience, including summer and fellowship jobs and consultant positions. Job titles are capitalized. The current job must have a location (city, state, country); previous positions may be listed without one. Do not abbreviate city names, Company, Laboratory, or Department. Use standard names for all countries. If there is space, information the author provides about previous publications may be included at the end of this paragraph. Edit out long lists of published books or articles. Instead use the sentence "s(he) is the author of several books and numerous published articles." The format for listing publishers of an author's books within the biography is: *Title of the Book* (publisher name, year) similar to a reference. (Note, use the word "titled" not "entitled" to introduce the book [e.g., He is the author of the book titled *Stochastic Analysis and Applications* (Taylor & Francis, 2012)]). List author affiliations with non-IEEE journals. Note IEEE TRANSACTION AND JOURNAL TITLES should be in small caps; *IEEE Magazine Titles* should be in italics; and non-IEEE titles should be in italics. List previous and current research interests. Do not repeat the author's name in the second paragraph; use the supplied pronoun.

Third Paragraph: The third paragraph begins with the author's title and last name (e.g., Dr. Smith, Prof. Jones, Mr. Kajor, Ms. Hunter). It lists the author's memberships in professional societies other than the IEEE and his or her status as a Professional Engineer if applicable. Finally, list awards and work for IEEE committees and publications, affiliation with other professional societies, and symposia.

Personal notes such as hobbies should not be included in the biography. Authors may include an external link to their work, this should appear as "For more information, see <http://website.of.author>" This should be the full URL and not an abbreviated link.

Examples:

Michael C. Author Jr. (Fellow, IEEE) was born in New York, NY, USA, in 1969. He received the B.S. degree in applied mathematics from the University of Michigan, Ann Arbor, MI, USA, in 1989, the M.S. degree in mathematical physics from Stanford University, Stanford, CA, USA, in 1991, and the Ph.D. degree in electrical engineering from the Massachusetts Institute of Technology, Cambridge, MA, USA, in 1995.

From 1993 to 1995, he was with Raytheon Corporation, Bedford, MA, USA. From 1995 to 1996, he was with the General Electric Space Laboratory, Valley Forge, PA, USA. From 1996 to 1997, he was a Fulbright Lecturer at the University of Madrid, Madrid, Spain. He is currently an Associate Professor of electrical engineering at the University of Maryland, College Park, MD, USA. His research has been concerned with reentry plasma effects and microwave diagnostics of plasmas.

Dr. Author is a Registered Professional Engineer in the State of Pennsylvania. For more information, see <http://website.of.author>.

Katsunari Okamoto was born in Hiroshima Prefecture, Japan, in 1949. He received the B.S. degree from Rutgers University, New Brunswick, NJ, USA, in 1979, and the M.S. degree from Monmouth University, Long Branch, NJ, USA, in 1984.

He was a Postdoctoral Fellow at the University of Tokyo, Japan, in 1978. He joined the Ibaraki Electrical Communication Laboratory, N.T.T., Ibaraki-ken, Japan, in 1979, where he was engaged in research on the optimum waveguide structure of optical fibers. At present, he is a Member of Technical Staff at Bellcore, Red Bank, NJ, USA.

Dr. Okamoto is a member of the Institute of Electronics and Communication Engineers of Japan.

Squibs

If the author chooses not to publish his/her biography and photograph, a squib is used. Example:

James A. Author (Fellow, IEEE), photograph and biography not available at the time of publication.

If *all* authors of the article opt not to publish his/her biography and photograph, no squib is used.

C. Other Text

Inclusive Language

To avoid the use of insensitive terms/phrases, please refer to the Inclusive Language Guide in the Appendix for replacement text. Use “people-first language,” i.e., the person has X; has been diagnosed with X; uses a X; etc. More information can be found [here](#).

Footnotes

Footnotes should be numbered in consecutive order throughout the text. Each footnote should be a new paragraph. The footnote numbers are superscripts in text and in the actual footnotes. In text, place the superscript footnote numbers after punctuation such as periods, commas, parentheses, and quotation marks, but generally before dashes, colons, and semicolons in a compound sentence. The footnotes should be placed at the bottom of the text column in which they are cited.

Lists in Text

There are three types of lists in text: run-in lists, displayed lists, and where lists. The ordering of labeling for all lists is 1), 2), 3) followed by a), b), c), and then i), ii), iii). Note the single (ending) parenthesis. The order of indentation is 1 em, 2 ems, 3 ems.

Run-In Lists: Lists that run in with text must be grammatically correct. They must also be introduced by a colon, separated by semicolons, and have parallel construction. Example:

The carrier–phonon interaction matrices are given by: 1) polar optical phonons; 2) deformation potential optical phonons; and 3) piezoelectric acoustic phonons.

Displayed Lists: Lists that are displayed may be either incomplete sentence items or full sentence items. Incomplete sentence items contain a few items, are very short, are grammatically parallel, and are handled in two ways. If the items are not mentioned in the text or are fewer than three items, run in as shown in the example for run-in lists. If, however, the items are mentioned later in the text, introduce the item with a colon, number the items, begin the entry with a lowercase letter, and set block paragraph style. Use semicolons between items and a period at the end of the list. Example:

This operating scenario provides all of the contributors necessary to configure a resonant power distribution system:

- 1) implementation of capacitor power factor correction on the power line;
- 2) presence of nonlinear load;
- 3) tuning of the power line by the load adjustments to a frequency present in the nonlinear generator.

Incomplete sentence items that are mentioned in text may also be formatted as shown in the example for full sentence items.

Example:

The three problems are related in the following sense:

- 1) Additional cost constraint;
- 2) Relaxation of the constraints is permitted;
- 3) Limited budget optimization is a general optimization problem.

Full sentence items may be introduced by “that” or other words taking object and end with a period. Number all items, start each entry with a capital letter, and end with a period. Example:

The synthesis is performed in three major steps.

- 1) Geometry is generated for the selected module variants.
- 2) Shape variants using different fold counts for resistors are generated for each module.
- 3) Routing and postprocessing complete the final layout.

Where Lists: Where lists define variables in the equations preceding the list. They are characterized by incomplete sentences and follow the same rules as *Nomenclature* lists, with the following exceptions.

- 1) There is no primary heading.
- 2) The left-hand side is indented one em space.
- 3) The first letter on the right-hand side is lowercase.
- 4) Each item ends with a semicolon (except for the last item, which ends with a period).
- 5) The lists are at least three items long; if fewer than three items, the list is generally run in paragraph form.

Example:

where

$$\Delta v_S = \Delta V_S \cos(\omega t + \phi');$$

ΔV_S amplitude of supply voltage flicker;

ω' angular frequency of supply voltage flicker;

V_{Sf} supply voltage amplitude;

ω supply angular frequency.

Note the alignment of the equal sign with the right-hand side.

Lists having mixed items (start with an incomplete item, then have a full sentence explanation) are treated as a full sentence item list.

Dedication Line(s)

Dedication lines are usually run on the first page of an article, immediately above the Abstract.

Example: *Dedicated to the work of J. W. Walters.*

Note Added in Proof

One may wish to add a brief note in the proof stage, citing results obtained after acceptance of the article or mentioning additional references that have come to their attention since the article was accepted. This added information is usually inserted at the end of the Conclusion section of the article or in whatever section contains the last paragraph of the main body of the article. As long as the note is not a major change to the article or more than a few lines long, the addition generally does not require further review procedures. Use the tertiary heading “Note Added in Proof.” (run into text), but set in boldface italic with no enumeration and an em space indent.

Examples:

Note Added in Proof: The author is an owner of the company which manufactured the tubes used in these experiments.

Note Added in Proof: Additional information about similar research can be found at www.newresearchresults.com.

D. Other Types of Papers

Editorials

This category of papers includes the various types of introductory papers, such as Editorials, Guest Editorials, Forewords, Introductions, and Editorial Announcements that appear at the beginning of issues as nontechnical introductory material. The Editorial may contain illustrations, citations, and references. Citations to articles in the issue should be listed as “Related Works” instead of in the reference section. It may contain a photograph and biography of each guest editor when it is a Guest Editorial for a special issue or section. An acknowledgment does

not contain a heading. *Note:* In the Editorial, the Acknowledgment does not need to be written in third person and there is no Abstract.

Title: An editorial may be titled simply “Editorial,” “Guest Editorial,” “Foreword,” “From the Editor,” etc. However, if the editorial includes its own title, it should be preceded by the article type and a colon.

Example:

Guest Editorial: Emerging Topics of DC-DC Converters for Solar PV

Byline: Note that the byline for the Editorial does NOT appear below the title as it does in a full-length article. The name of the author of the Editorial or Foreword (usually the Editor or Guest Editor) (called “signature”) appears at the end of the Editorial.

Example:

MARVIN K. SAIN, *Guest Editor*
Department of Electrical Engineering
University of Illinois
Urbana, IL 60617 USA

Brief Papers

These articles contain Abstracts and an initial cap. The byline includes the membership grade. They do not contain biographies and photographs of the authors.

Short Papers, Letters, Correspondence, and Communications

Short papers are set up like full-length articles. The membership grade is not included in the byline. Author biographies and photographs are not included. Footnotes, captions, and references may be included.

Letters are a type of short paper that have a strict low page limit and appear at a back section of an issue. Note that these letters are not the same as research letters formatted as regular papers without biographies that make up entire volumes or issues (e.g., IEEE Antennas and Wireless Propagation Letters, IEEE Electron Device Letters, etc.).

Correspondence and communications also use the short paper format, but are typically only a few paragraphs in length. These include letters to the editor.

Comments and Replies

Comments are generally in response to a previously published article. The Comments and Author(s) Reply are short papers published together in that the “Reply” is in response to the Comments. These short items must include Abstracts and Index Terms. A special format applies for Comments and Author(s) Reply. Begin the first sentence with “In the above article [1], ...” Reference [1] is the commented article’s citation and will appear as Reference [1] in the References section.

Some publications refer to these articles as Discussions and Closures.

Example of the Comments:

Title: Comments on “Harmonics: The Effects on Power Quality and Transformers”

Byline: Keith H. Sueker

Footnote:

Received 15 July 2006.

The author is with the School of Engineering, Vanderbilt University, Nashville, TN 37235 USA (e-mail: k.sueker@ieee.org).

Digital Object Identifier 10.1109/JQE.2006.12345

NOTE: The footnote here relates back to the original article being commented upon. The title is not repeated.

Example of the Reply:

Title: Authors' Reply

Byline: Robert D. Henderson and Patrick J. Rose

Footnote:

Received 3 October 2006; accepted 5 October 2006. Date of publication 2 November 2006; date of current version 25 November 2006.

The authors are with RDH Consultants, Inc., Charlotte, NC 28241 USA (e-mail: corresponding@author.com).

Digital Object Identifier 10.1109/JQE.2006.12348

Corrections/Errata/Addendums

The format for Corrections, Errata, and Addendums is basically the same as for the Comments, except that Corrections/Errata/Addendums do not carry a Reply. Abstracts and Index Terms are optional. However, Abstracts and Index Terms should either both be present or both be absent. All run a copyright line. A *Correction* is a notice that makes note of an error by the author in their original writing. An *Erratum* is a notice of an error introduced by the publisher. An Addendum is an additional, short statement that relates to/supplements the published article. It should follow the standard format of a Correspondence.

Note: The plural form of the word is used in the title, even if there may be only one correction. All Corrections/Errata *must* carry the byline as the same form as the original article; this ensures that the two articles will be linked properly. IEEE will not republish the original article with the changes made; only the correction/erratum detailing the error(s) and resolution(s).

Example of a "Corrections" article:

Title: Corrections to "On the Exact Realization of LOG-Domain Elliptic Filters Using the Signal Flow Graph Approach"

Byline: Costas Psychalinos and Spiridon Vlassis

Footnote:

Received 1 May 2003.

The authors are with the Physics Department, Electronics Laboratory, Aristotle University of Thessaloniki, GR-54124 Thessaloniki, Greece (e-mail: cpsychal@physics.auth.gr; svals@skiathos.physics.auth.gr).

Digital Object Identifier 10.1109/TCSII.2003.814788

Example of an Erratum:

Title: Erratum to "Harmonics: The Effects on Power Quality and Transformers"

Byline: Robert D. Henderson and Patrick J. Rose

Footnote:

Received 20 January 2004.

The authors are with RDH Consultants, Inc., Charlotte, NC 28241 USA (e-mail: pjrose@rdh.com).

Digital Object Identifier 10.1109/TVLSI.2004.830244

Example of an Addendum:

Title: Addendum to "Harmonics: The Effects on Power Quality and Transformers"

Byline: Robert D. Henderson and Patrick J. Rose

Footnote:

Received 20 January 2004.

The authors are with RDH Consultants, Inc., Charlotte, NC 28241 USA (e-mail: pjrose@rdh.com).

Digital Object Identifier 10.1109/TVLSI.2004.830244

Book Reviews

Some publications carry Book Reviews. They are the same as a short paper or correspondence; however, the title runs additional information about the book that is being reviewed. The title is separated from the book's author by an em dash. Included in parentheses is the city of publication, publisher, date of publication, the total number of

pages of the book, and the price. Outside of the parentheses is the reviewer's name in italics. Some Transactions carry a short biography of the reviewer under the title. Book Reviews appear in the table of contents with a listing for both the author of the book and the reviewer. Example:

Title and Byline:

The Analysis and Design of Pneumatic Systems—B. L. Andersen. (New York: Wiley, 1987, 302 pp., \$65.00.)
Reviewed by J. L. Shearer.

First Footnote:

The reviewer is with the College of Engineering, Idaho State University, Pocatello, ID 83209 USA. Digital Identifier 0090-6778/TNN.2005.828433.

Table of Contents:

The Analysis and Design of Pneumatic Systems—B. L. Andersen*Reviewed by J. L. Shearer* 123

Obituaries/In Memoriam

Obituaries are usually run as the first page of an issue, like an Editorial. They are set up with the same specs as Editorials.

E. Writing Style for Transactions

The following provides a summary of the most important style distinctions to be made in the writing of a Transactions article.

Acronyms

Define acronyms the first time they appear in the Abstract as well as the first time they appear in the body of the article, written out first as part of the sentence, followed by the acronym in parentheses. Widely used or familiar terms should be defined (see the Common Acronyms and Abbreviations list in the Appendix for some terms that must be defined the first time they are used in text). Acronyms do not need to be defined in the text if mentioned in the Nomenclature. Coined plurals or plurals of acronyms do not take the apostrophe as per *Chicago Manual of Style*. Example: FET (singular); FETs (plural).

Indefinite articles are assigned to abbreviations to fit the sound of the first letter: an FCC regulation; a BRI.

Spelling

Note that IEEE Transactions use the first spelling of a word as given in the main entry of *The Merriam-Webster Dictionary*.

British Spellings and Terminology: Change all British spellings to American spellings. In particular, watch for “our” endings in words like “behaviour” (change to “behavior”) and “re” endings in words like “centre” (change to “center”). Also watch for the use of “s” rather than “z” in words like “polarisation” (change to “polarization”). See “Common Hyphenations and Misspellings” in the Appendix.

Trademarks

The trademark symbols TM and ® are no longer used. Capitalize the first letter in the trademark name only. The symbols TM and ®, which often accompany registered trademark names on product packaging and in advertisements, need not be used in running text. Optionally, for the first occurrence of a trademarked product, a footnote superscript can be placed after the trademarked name, with a matching footnote that reads “Trademarked.” or “Registered trademark.”

Plurals

Plurals of units of measure take the “s.” For example, the plural form of 3 mil is 3 mils; 3 bits/s instead of 3 bit/s. The plural of calendar years do not take the apostrophe before the “s.” For example, the plural form of 1990 is 1990s.

Hyphenation Rules

For hyphenation and spelling guidelines, IEEE style follows: 1) the list of preferred spellings and hyphenated words can be found in the Appendix; 2) the guidelines discussed in the Grammar and Usage in Transactions section of this guide; and 3) the first version of the spelling given in the most recent edition of *The Merriam-Webster Dictionary*. Do not hyphenate most compound modifiers if they occur after the noun being modified, even if hyphenating them before the noun.

Examples:

The plan was well prepared. The man was little known. The woman was better qualified. His boat was 42 feet long. He has a 42-foot-long boat. T was the data period of the 40-Gb/s data signal. The 160-GHz MLLD was a diode in which a 40-nm-long saturable absorber was located.

NOTE: Do not use the *IEEE Standards Dictionary* for hyphenation guidelines as no attempt is made there for consistency in hyphenation. The *Standards Dictionary* is quite useful for its definitions and acronyms list in its back section.

The most important hyphenation guideline is to be certain that the hyphenation for a particular word or group of adjectives is consistent within a particular article.

The En or Em Dash

The en dash represents the words “to,” “through,” or “and.” Use it between page numbers, figure citations, academic years, proper nouns, names, a range of values, or for opposites.

Examples:

- pp. 10–15,
- 1984–1990,
- Jones–Smith theorem,
- input–output,
- voltage–current curve,
- analog–digital converter,
- 10–20 cm.

Also, use the en dash in chemical abbreviations such as Ni–Al–Si. When using the en dash to represent a range, if the word “from” occurs, the word “to” must be used rather than an en dash (e.g., ranges from 5 to 50 times).

The em dash is used in ordinary writing to mark a suspension of the sense. It is also used like parentheses, to mark a subordinate thought within a sentence.

Grammar

Check closely for lapses of clarity, subject/verb agreement, and parallel clause construction. See the following examples:

Number:

A number of samples were taken ...

A number *N* expressing the relation *x/y* is chosen ...

Data:

The data were collected ... (always plural)

Series:

A series of tests was run ... (always singular with “a”)

Some, All, Half:

Some (all, half) of it is ...

Some of them are ...

For example:

Use “all of” with another pronoun, such as “these” or “those,” and before singular nouns. For collective and plural nouns, use “all.”

Quantity:

Three volts were applied ...

Four grams were added ...

Contractions

Contractions such as “don’t” and “can’t” are not used in technical text. Change to “do not” and “cannot.”

Note: “don’t care,” “best-case,” and “worst-case” are allowed and used often in journals like TCAD.

Capitalization

In general, discourage capitalization in text except where absolutely necessary. For example, only proper names attached to the names of laws, principles, theorems, etc., get capitalized (Abel’s theorem, Newton’s first law, etc.). Computer commands are in computer tags and remain small caps; most computer languages (Cobol, Java, LISP, PERL, etc.) are upper and lowercase. Earth should be capitalized when referring to the planet.

Dates

Use the international date format for all dates in the article. Spell out the month. (Note: This does not include references. Continue to follow IEEE Reference Style.)

4 June 2002

23-31 October 2019

3 November 2021–4 December 2021

Percentages and Decimals

Always use the number and the percent sign when dealing with percentages. The percentage symbol is repeated in lists and ranges.

Only 2% of the transformers failed the test.

The students made up 20%-30% of the population.

When using decimal fractions in text, include the zero before decimal if needed for clarity, otherwise omit it. Do not include the zero(s) after the last digit following a decimal:

.25

0.8

Ranges With Units

When reporting ranges, there should be no unit after each number unless the units are different:

40–50 mm

50 inches to 7 feet

2 × 5 cm

Math

Some brief guidelines for writing math are explained here.

- 1) Variables are set italic; vectors are usually boldface italic.
- 2) Remove commas around variables in text.
- 3) Always add a zero before decimals, but do not add after (e.g., 0.25).
- 4) Check the use of the parentheses and brackets i.e., [0,1).

- 5) Spell out units used in text without quantities (e.g., “where the noise is given in decibels”). For units appearing with quantities, use the standard abbreviations listed in the Table of Units and Quantity Symbols in the Appendix, and units used as compound adjectives may be hyphenated only if needed for clarity: 10-kV voltage, 5-in-thick glass. Do not insert a hyphen when they are not used as adjectives: a current of 2 A, a line 4 in long, a length of 3.05 mm.
- 6) Always use a regular space and not a thin space between numbers and units in text.
- 7) Use thin spaces instead of commas between numbers in tens or hundreds of thousands (e.g., 62 000, 100 000, but 4000).
- 8) Always make sure μ is μm , “micron” is “micrometer,” “submicron” is submicrometer.” Always change cycle per second to hertz (Hz); cycle per second may not appear as cycle, cps, c/s, csec.
- 9) In text, fractions may be broken down (shilled) multiline (built-up) so they can be placed on one line. Sometimes parentheses may need to be added to distinguish between expressions, especially when a minus

appears [e.g., $\frac{a}{b-c}$ becomes $a/(b-c)$], $\frac{c-d}{k+4}$ becomes $[(c-d)/(k+4)]$. This may be done to save space, but is not a necessity.

- 10) In exponential expressions [e.g., $e^{-(j\omega t)xyzk}$], there are sometimes long and complicated superscripts. These may be brought down in line with the substitution of “exp” for “e” and the addition of square brackets (e.g., $\exp[-(j\omega t)xyzk]$).
- 11) Distinguish between lowercase italic “ell” or “oh” versus one and zero.
- 12) Always use numerals for numbers written with units. Otherwise, spell out numbers below 11, and use numerals for others unless they begin a sentence or are combined in a phrase (gives 7 to 13 times more).
- 13) Use zeroth, first, n th, $(k+1)$ th, not 1st, 2nd, $(k+1)$ st, etc.
- 14) Use the word “Equation” at the start of a sentence, but in text, just use the number [e.g., in (1)].
- 15) Use the \$ symbol versus “dollars” in sums of money.
- 16) The slash (/) is acceptable in place of the word “per” when it lends to the clarity of the sentence. For example: “the ratio of 16 samples/s to 35 samples/s as compared to ...”

Ellipses: In mathematics, you may use dots (ellipses) to show continuation in an expression (e.g., $x_2, \dots, x_{16}$). The type of mathematical expression will determine whether the ellipses points are set on the baseline or centered. If commas or operational signs are present, they are placed after each term and after the three ellipses points. If operational signs are used, the ellipses are centered on the operator. When commas are used, the ellipses are on the baseline. Example:

$x_1, x_2, \dots, x_n$ not $x_1, x_2 \dots x_n$
 $x_1 + x_2 + \dots + x_n$ not $x_1 + x_2 + \dots x_n$
 $y = 0, 1, 2, \dots$ not $y = 0, 1, 2 \dots$
 $x_1 x_2 \dots a_n$ not $x_1 x_2 \dots a_n$

Conditions: In displayed equations, a comma or parentheses and a two-em space is inserted between the main expression and the condition following it. Example:

$$x = yn^{-2} \quad \forall n = 3$$

$$x = yn^{-2}, \quad \text{if } n = 3 - y^{-4}.$$

$$x = yn^{-2}, \quad y = 3, \quad m$$

NOTE: There is no comma before a for all “ $\forall$ ” symbol.

Compound Units: Compound units should be separated by a center dot (e.g., 4 V · s), but a slash may be used since this has a different meaning (for instance, 6 V/s means volts per second). It is also possible to use a negative power to put a unit in the denominator: $\text{cm/s}^2 = \text{cm} \cdot \text{s}^{-2}$. Parentheses may be used to clarify a unit: $\text{g}/(\text{cm} \cdot \text{s})$ or $\text{g} \cdot \text{cm}^{-1} \cdot \text{s}^{-1}$.

Use of Periods and Commas: Equations which conclude a sentence should end with a period. The only time punctuation is used to lead into an equation is when the lead-in text is a complete sentence. Example:

where we had the following:

$$x = Y + Z.$$

or where, i.e.,

$$x = Y + Z.$$

Commas appearing at the ends of equations are deleted unless they are critical to the punctuation of the sentence containing the equation.

Equation Numbers

Equation numbering should be consecutive, should appear flush right on line with the last line of an equation, should not have repeats or missing numbers, and should use a correct numbering style.

Displayed Equations

Material in displayed equations is automatically italic unless you indicate otherwise. Some simple general rules apply. All variables are italic. Function names and abbreviations are Roman, as are units, unit abbreviations, complete words, and abbreviations of words. Superscripts and subscripts follow this same formula: when they are variables, they are italic; when they are abbreviations of words (such as “in” and “out” for input and output), they are Roman. Single-letter superscripts and subscripts may be italic even if they are abbreviations, unless this leads to inconsistency between italic and Roman characters for similar types of subscripts.

F. General Layout Rules

- 1) Figures and tables are placed at the tops of columns as close to their first mention as possible, but preferably after the mention.
- 2) Figures and tables progress vertically, not horizontally, on pages.
- 3) Footnotes must appear at the bottom of the column where they are first mentioned.

III. GRAMMAR AND USAGE IN TRANSACTIONS

A. Rules of Grammar

The principles of style below focus on fundamentals of modern usage. Particular emphasis is given to the rules most commonly violated.

- 1) **Form the possessive singular of nouns by adding “s”** (*Avogadro’s theorem*). Follow this rule unless the final consonant is an s (*Burns’ theorem*). Possessive pronouns (*hers, its, yours, theirs, ours*) have no apostrophe. Indefinite pronouns use the apostrophe to show possession (*someone’s rule*). Contractions use an apostrophe (*it’s for ...; it is*). Possessives do not (*its losses*).
- 2) **In a series of three or more terms, use a comma immediately before the coordinating conjunction** (*usually and, or, or nor*).
- 3) **Enclose parenthetic expressions between commas** (*Improvement, as shown in Fig. 1, is attained by the addition of the cogeneration*). Brief phrases or single words, such as *however*, may or may not be parenthetic (such connectives at the head of a sentence are more commonly left unpunctuated). The commas may be omitted if the interruption to the flow of the sentence is slight. In this case, never omit one comma and leave the other. Remember that many seemingly single commas stand for a pair. Clauses or phrases at the beginning or end of sentences do not look parenthetical, but often they might just as well be placed in the middle, in which case they would be found punctuated at both ends. At the beginning of a sentence, such an element is set off by what should be thought of as the second comma in a pair. For instance, note the three possible positions illustrating a parenthetical element of this kind: *However the sum may later change, it is calculated now/The sum is calculated now, however it may later change/The sum, however it may later change, is calculated now*. In all three examples, the meaning remains constant; the single commas of the first and second sentences have the same parenthetical function as the paired commas of the third.

Parenthetic material such as dates take the comma(s) as follows: *14 February 1996* or *April to June 1996* or *Saturday, 9 March 1996*.

The abbreviations etc., i.e., and e.g., are parenthetic and use the comma as follows: cables, transformers, etc., are needed. Abbreviations for academic degrees, titles following a name, and certain restrictive terms of identification should be punctuated as follows:

Robert D. Lorenz, Ph.D.

Ian T. Wallace, Member, requests that...

E. A. Brockmann Jr. states that...

Restrictive clauses are not parenthetic and are not set off by commas: *The proof that (or which) (restrictive clause should be “that” while nonrestrictive is “which”; “who” can be restrictive or nonrestrictive, depending on how it is used) is given in this section is not complete.*

Nonrestrictive clauses are parenthetic and are set off by commas: *The address i, which is the starting address of the message, is then transferred to a queue list on the processing part ...*

The nonrestrictive clause always takes “which” and is surrounded by commas. The restrictive clause can take “that” or “which”; “that” is preferred.

- 4) **A semicolon is used to link two independent clauses with no connecting words. You can also use a semicolon to join two independent clauses together with one of the following conjunctive adverbs: however, moreover, therefore, consequently, otherwise, nevertheless, thus, etc.**
- 5) **Use a colon after an independent clause to introduce a list.**
- 6) **Punctuation always goes inside quotation marks, except for the colon and semicolon.** Use single quotation marks around quotes within quotes. Quotes may be used around a new or special usage of a term the first time only, but use of quotes in this manner should be kept to a minimum.
- 7) **Direct quotes should be set in quotation marks in roman font.** Text should not be in italics.
- 8) **Do not use double parentheses in text expressions, but keep them in math.** For example, (see (10)) should become [see (10)].
- 9) **All acronyms and numerical plurals do not use apostrophes**, i.e., *FETs, 1980s* (Note: Some exceptions may apply in mathematical writing.)
- 10) **Compound nouns made from a one-syllable verb and a short adverb are one word when found that way in the dictionary** (setup, takeoff, breakup). Compound nouns are likely to be two words, without a hyphen, or one word (bandwidth, bypass, flowchart, phase shift, sideband, standing wave). Compound nouns of more than two words can be hyphenated.

- 11) **A pair of words, modifying a third word separately, does not get a hyphen** (a tall water tower, a hot metal cylinder). If the first word modifies the second, and the pair together modify the third, there is a hyphen between the pair (a highfrequency signal, a secondorder equation). The exception to this is the adverb ending in “ly,” which needs no hyphen to join it to the next word.
- 12) **A hyphen is not used after the comparative or the superlative** (a higher order equation, a worst case value, nearest neighbor method). Do not hyphenate chemical compounds (sodium chloride crystals). Alloys and mixtures take the en dash (Ni–Co, He–Ne laser).
- 13) **Do not use commas between adjectives** (a planar equiangular spiral antenna).
- 14) **Do not hyphenate predicate adjectives** (... is well known, ...is second order).
- 15) If you are unsure, check *The Merriam-Webster Dictionary* to see if words are hyphenated.
- 16) **Compound verbs are generally hyphenated** (arc-weld, freeze-dry). Keep the hyphen when using the participles of such verbs as adjectives (freezedried, arcwelded). However, verbs with up, out, down, off, on, etc., do not have a hyphen, although the nouns formed from them may be hyphenated or one word (verb: set up, break down, read out; noun: setup, breakdown, readout).

Words Often Confused

Affect: to change or modify (verb).
Effect: result (noun); cause (verb).

Alternate: a substitute.
Alternative: a matter of choice.

Among: involves more than two things.
Between: involves more than two things, but considers each individually.

Compare to: point out resemblances between different objects.
Compare with: point out similarities and differences between same objects.

Compose: to make up or form: a set composed of members.
Comprise: to be made up of; to be formed by: a set comprising members; members comprising a set.

Farther: distance.
Further: quantity.

Fewer: modifies plural nouns specifying countable units, e.g., fewer tubes.
Less: modifies singular mass nouns and singular abstract nouns, e.g., less air.

Imply: something suggested though not expressed.
Infer: something deduced from evidence.

Number: used when objects can be counted: a large number of people.
Amount: used when objects cannot be counted: a large amount of water.

Principal: chief, main, most important (adjective).
Principle: a rule (noun).

Precede: come before.
Proceed: continue, advance.

That: (defining, restrictive).
Which: (nondefining, nonrestrictive)

IV. APPENDIX

A. Some Common Acronyms and Abbreviations

NOTE: Asterisks (*) indicate terms which must be defined the first time they are used in text. Other terms listed here may be used without definition.

#	
1-D	one-dimensional
2-D	two-dimensional
3-D	three-dimensional
4-D	four-dimensional
A	
ac	alternating current
A–D, A/D	analog-to-digital
AF	audio frequency*
AFC	automatic frequency control*
AGC	automatic gain control*
AM	amplitude modulation
APD	avalanche photodiode
AR	antireflection*
ARMA	autoregressive moving average*
ASIC	application-specified integrated circuit*
ASK	amplitude shift keying
ATM	asynchronous transfer mode
av	average (subscript)*
avg	average (function)
AWGN	additive white Gaussian noise*
B	
B–E	base–emitter source
BER	bit error rate*
BPSK	binary phase-shift keying
BWO	backward-wave oscillator*
C	
c.c.	complex conjugate (in equations)
CCD	charge-coupled device*
CDMA	code division multiple access*
CD-ROM	compact disk read-only memory
CIM	computer integrated manufacturing*
CIR	carrier-to-interference ratio*
CMOS	complimentary metal–oxide–semiconductor
CPFSK	continuous phase frequency-shift keying*
CPM	continuous phase modulation*
CPSK	continuous phase-shift keying*
CPU	central processing unit
CRT	cathode-ray tube
CT	current transformer*
CV	capacitance–voltage
CW	continuous wave*
D	
dc	direct current
DC	directional coupler

DF	direction finder*; deuterium fluoride; degree of freedom*
DFT	discrete Fourier transform*
DMA	direct memory access*
DPCM	differential pulse code modulation*
DPSK	differential phase-shift keying*
E	
EDP	electronic data processing
EHF	extremely high frequency*
ELF	extremely low frequency*
EMC	electromagnetic compatibility*
EMF	electromotive force*
EMI	electromagnetic interference*
ems	expected value of mean square*
F	
FDM	frequency division multiplexing*
FDMA	frequency division multiple access*
FET	field-effect transistor
FFT	fast Fourier transform*
FIR	finite-impulse response*
FM	frequency modulation
FSK	frequency-shift keying*
FTP	file transfer protocol
FWHM	full-width at half-maximum*
G	
GUI	graphical user interface
H	
HBT	heterojunction bipolar transistor
HEMT	high-electron mobility transistor
HF	high frequency
HTML	hypertext markup language
HV	high voltage
HVdc	high voltage direct current
I	
IC	impedance compensation*; integrated circuit
ID	inside diameter; induced draft*; interdigital*
IDP	integrated data processing*
IF	intermediate frequency
IGFET	insulated-gate field-effect transistor
i.i.d.	independent identically distributed*
IM	intermediate modulation
IMPATT	impact ionization avalanche transit time (diode)
I/O, I-O	input-output
<i>IR</i>	infrared
IR	current-resistance
ISI	intersymbol interference
<i>I-V</i>	current-voltage
J	
JFET	junction field-effect transistor
JPEG	Joint Photographers Expert Group
L	
LAN	local area network
<i>LC</i>	inductance-capacitance
LED	light-emitting diode

LHS	left-hand side*
$L-I$	light output–current
LMS	least mean square
LO	local oscillator*
LP	linear programming*
LPE	liquid phase epitaxy*
LR	inductance–resistance
M	
MESFET	metal–semiconductor field-effect transistor
MF	medium frequency*
MFSK	minimum frequency-shift keying
MHD	magnetohydrodynamics
MIS	metal–insulator–semiconductor
MLE	maximum-likelihood estimator*
MLSE	maximum-likelihood sequence estimator*
MMF	magnetomotive force
MMIC	monolithic microwave integrated circuit*
MoM	method of moments*
MOS	metal–oxide–semiconductor
MOSFET	metal–oxide–semiconductor field-effect transistor
MOST	metal–oxide–semiconductor transistor
MPEG	Motion Pictures Expert Group
N	
NA	numerical aperture*
NIR	near infrared response*
NMR	nuclear magnetic resonance*
n-p-n	(diode)
NRZ	nonreturn to zero*
O	
OD	outside diameter
OEIC	optoelectronic integrated circuit*
OOP	object-oriented programming
P	
PAM	pulse-amplitude modulation*
PC	personal computer
PCM	pulse-code modulation*
pdf	probability density function*
PDM	pulse-duration modulation*
PF	power factor*
PID	Proportional-integral differential
p-i-n, p-n-p	(diode)
PLL	phase-locked loop*
PM	phase modulation*
PML	perfectly matched layer
pp, p–p	peak-to-peak*
PPM	pulse-position modulation*
PRF	pulse-repetition frequency*
PRR	pulse-repetition rate*
PSK	phase-shift keying*
PTM	pulse–time modulation
p.u.	per unit*
PWM	pulse width modulation*
Q	

Q	quality factor; figure of merit
QoS	quality of service
QPSK	quaternary phase-shift keying
R	
RAM	random access memory
RC	resistance–capacitance
R&D	research and development
RF	radio frequency
RFI	radio frequency interference*
RHS	right-hand side*
RIN	relative intensity noise*
RL	resistance–inductance
rms	root mean square
ROM	read-only memory
RV	random variable
S	
SAW	surface acoustic wave*
SGML	standard generalized markup language
SHF	super high frequency*
SI	International System of Units; severity index*
SIR	signal-to-interference ratio
S/N , SNR	signal-to-noise ratio
SOC	system-on-a-chip*
SSB	single sideband*
SW	short wave*
SWR	standing-wave ratio*
TDM	time-division modulation*; time-division multiplexing*
TDMA	time-division multiple access*
TE	transverse electric
TEM	transverse electromagnetic
TFT	thin-film transistor*
TM	transverse magnetic
TVI	television interference*
TWA	traveling-wave amplifier*
U	
UHF	ultrahigh frequency
UV	Ultraviolet
V	
VCO	voltage-controlled oscillator*
VHF	very high frequency*
$V-I$	voltage–current
VLF	very low frequency*
VLSI	very large scale integration*
W	
WAN	wide area network
WDM	wavelength division multiplexing*

B. Common Hyphenations and Misspellings

A	Boltzmann's constant	Doppler
a posteriori	Boolean algebra	drain–source [en dash]
a priori	broadband	dropout
Abelian	bulk–source [en dash]	dyadic
accommodate	bus (not buss)	E
acknowledgment	bypass	eccentricity
acoustoelectric	C	eigenfunction
acoustooptical	Cartesian	eigenvalue
ad hoc	Cascade	eigenvector
ad hoc networks	cascode	elastance
adder	Cauchy's inequality	elasto-optical
aerospace	Chebyshev	electro-optic
aftereffect	(not Tchebbycheff)	elliptical coordinates
airborne	chi-square	elliptic integrals
all-pass (adj)	Clebsch–Gordan coefficient	emitter–bulk [en dash]
Alnico	coauthor (also, coworker)	end-effector
alphameric	coax (coaxial)	endfire
alphanumeric	collinear (not colinear)	endpoint
analog (not analogue)	continuous-time (adj)	et al.
appendixes	coset	Euler function
arc-back (n, adj)	costate	exponentiate
arc-over (n, adj)	Coulomb wave function	F
axle	counterclockwise	fan-in
B	counterexample	fan-out
back EMF	coworker	far-field (adj)
back-end (adj)	coupled-mode (adj)	fast Fourier transform
backscatter	cross correlation	feedback
[band-letter(s)]-band (adj.)	crossover	feedback-free (adj)
(e.g., <i>X</i> -band)	cross section	first-order (adj)
band-limited (adj)	cross-sectional (adj)	flat-band
bandpass	crosstalk	flip-flop
band-shared (adj)	cutoff	flowchart
bandwidth	cybersecurity	flowmeter
bang-bang	D	flowthrough
base-emitter [en dash]	database	fold (twofold, <i>n</i> -fold)
base-collector [en dash]	deadtime (or dead time)	foreword
baseband	debug, debugged	formulas (not formulae)
baseline	Debye temperature	forward scatter
Bayes' rule	Dewar	4-vector
beamwidth	diagramed	front-end (adj)
Bernoulli polynomial	dielectric	Fresnel
Bessel function	diesel	G
bimetallic	digamma function	gate–source [en dash]
biomedical	Dirac	gate–drain [en dash]
blackbody	discretization	gauge (not gage)
	discusser	Gaussian distribution

Gegenbauer
gimbale
gradient
(the) Green's function
Gudermannian
H
half-angle
half-plane
half-space
half-wave
halfway
Hankel function
Heaviside
Hermite
Hermitian
Hertzian
higher order (adj)
high-order (adj)
high-pass (adj)
hookup
hydroelectric
I
iff (if and only if)
imbalance (n)
inasmuch as
indexes (plural of index)
indices (plural used in math)
infrared
inhomogeneous
input, inputted
input–output [en dash]
in situ
insofar as
in vitro
in vivo
integer
integral
integrand
integrator
integro-differential
Internet
Itô
J
Jacobian
Jacobi's polynomials
K
Kronecker delta
L
Lagrange

Lagrangian
Laguerre polynomial
Lame's transform
Laplace transform
Laplacian
Laurent series
left-hand side
leftmost
Legendre
Leibnitz (or Leibniz)
leveled
lightweight
like (suffix, close up)
line shape
lineup
linewidth
lockout
log-likelihood (adj)
lookup table
loudspeaker
lower order (adj)
low-order (adj)
low-pass (adj)
Lur'e
Lurie
Lyapunov (not Liapunov)
M
macro (noun)
magnetohydrodynamics
magneto optic
main lobe
makeup
manhole
man-hour
man-made
manpower
Markov process
<i>m</i> -ary
Mathieu's equation
matrices
mean-square
mid (prefix) close up
midband
midline
midplane
midpoint
miniscule
missile
modem

modulo (mod)
modulus
monotonic
monotonically
monotonicity
Mossbauer
<i>m</i> -sequence (noun)
multi (prefix) usually one word
multithreshold
Mylar
N
narrowband (adj)
<i>n</i> -ary
nearby
near-field (adj)
neoprene
Neumann
n-junction
n-layer
non (prefix) one word
non-Euclidean
non-Gaussian
non-Hermitian
nonnegative
non-Stokes'
nonzero
NP-hard
<i>n</i> th-order (adj)
<i>n</i> -tuple
<i>n</i> -type
<i>n</i> -well
O
ohmmeter
one-dimensional (adj)
ORED, ORing
ON-OFF
output, outputted
overall (adj)
P
parameterization
particle
passband
percent
Permalloy
Perspex
phaselength
phase shift
phasewidth

photoelectric
photoetch
photoresist
pickup
piecewise linear
piezoelectricity
p-i-n
pinchoff
p-junction
Planck's constant
p-n junction
p-n-p (not PNP)
$p^{+}\text{-}n\text{-}p^{++}$
Poisson distribution
positive definite
postmultiplication
pothead
potline
powerhouse
power plant
preceding
premultiplication
printout
proceeding
programmed
proof (suffix) one word
propagation
pseudo (prefix) one word
pseudorandom
p-type
pull-in
pull-out
pulselength
pulse shape
pulsewidth
punchthrough
p-well
Q
quadratic
quarter-wave
quartic
quasi- (prefix) hyphen
quaternary
Q value
R
radioactive
radio-astronomic
radio astronomy
radio frequency

random access (adj)
readback
READ head
readin (noun)
readout (noun)
real-valued (adj)
reentry
reexamine
Riccati
Riemann
right-hand side
rise time
root-mean-square (adj)
roundoff (adj)
Runge–Kutta
S
saddle point
scalar (magnitude)
scaler (machine)
scalor (rare)
self- (prefix) hyphen
self-adjoint
semi (prefix) usually one word
semi-infinite
servo (servomechanism)
servo amplifier
-shaped (hyphen)
sideband
sidelobe
signaling
slip ring
slow wave
so-called
solid-state (adj)
space-time
special-purpose (adj)
spirule
state of the art (noun)
state-variable (adj)
step-down
step-up
Stirling numbers
Stokes'
stopband
straightforward
strain gauge
Struve's function
Sturm–Liouville [en dash]

suboptimum
subproblem
succeeding
successive
summable, asummable
supercoding
supermartingale
supersede
switchgear
switchyard
T
table lookup
takeoff
Taylor expansion
Tchebyscheff (use Chebyshev)
Teflon
Teletype
teletypewriter
tensor
thin-film (adj)
threefold
3-space
throughput
time dependence
time-varying (adj)
tradeoff
traveling
two-port (or 2-port)
two's complement
-type (hyphen)
U
ultrahigh frequency
ultrasonic
ultraviolet
unbalance (verb)
V
Van de Graaf
van der Waals
vector
versus
vertical
vertices
W
watthour meter
wattmeter
waveband
waveform
wavefront

wave function
waveguide
wavelength
wavenumber
wave shape
wave vector
wideband
wide-sense (adj)
widespread
wise (suffix) one word
worldwide
worst case (adj)
WRITE head

X
x-axis
x-direction
X-ray (adj)
xy plane
Y
Yagi
Z
Zener diode
zero-input (adj)
zero-sum (adj)
zeroth-order (adj)
z transform

C. Table of Units and Quantity Symbols

NOTE: Asterisks (*) indicate SI units, preferred multiples of SI units, or other units acceptable for use with SI.

Unit	Unit Symbol	Sometimes Occur as: (do not use)	Applications and Notes	Quantity Symbol (for use as variables, etc.)
*ampere	A	amp, a	SI unit of electric current.	I U F
ampere-hour	Ah	amp-hr	Also A · h.	
*ampere (turn)	A	At	SI unit of magnetomotive force.	F
*ampere per meter	A/m		SI unit of magnetic field strength.	A H
ångström	Å	Å	Å $\triangleq 10^{-10}$ m. Deprecated (see ANSI/IEEE Std 268-1992).	
atmosphere, standard	atm		atm $\triangleq 101\,325$ Pa. Deprecated (see ANSI/IEEE Std 268-1992).	
atmosphere, technical	at		at $\triangleq \text{kgf/cm}^2$. Deprecated (see ANSI/IEEE Std 268-1992).	
*atomic mass unit (unified)	u		The (unified) atomic mass unit is defined as one-twelfth of the mass of an atom of the carbon-12 nuclide. Use of the old atomic mass unit (amu), defined by reference to oxygen, is deprecated.	
*atto	a		SI prefix for 10^{-18} .	
*attoampere	aA			
bar	bar	b, barye	bar $\triangleq 100$ kPa. Use of the bar is strongly discouraged (see ANSI/IEEE Std 268-1992). Except for limited use in meteorology.	
barn	b		b $\triangleq 10^{-28}$ m ² .	

barrel	bbl		bbl = 42 gal _{US} = 158.99 L. This is the standard barrel used for petroleum and petroleum products. Different standard barrels are used for other commodities.	
barrel per day	bbl/d			
baud	Bd	baud (w/prefix)	In telecommunications, a unit of signaling speed equal to one element per second. The signaling speed in bauds is equal to the reciprocal of the signal element length in seconds.	1/ τ
bel	B	b		
*becquerel	Bq		SI unit of activity of a radionuclide.	
billion electronvolts	GeV	bev, BeV	The name <i>gigaelectronvolt</i> is preferred for this unit.	
bit	b		In information theory, the bit is a unit of information content equal to the information content of a message, the <i>a priori</i> probability of which is one-half. In computer science, the name bit is used as a short form of <i>binary digit</i> .	
bit per second	b/s			
British thermal unit	Btu			
byte	B		A byte is a string of bits, usually eight bits long, operated on as a unit. A byte is capable of holding one character set.	
calorie (International Table)	cal _{IT}		Δ cal _{IT} 4.1868 J. Deprecated (see ANSI/IEEE Std 268-1992).	
calorie (thermochemical)	cal		Δ cal 4.1840 J. Deprecated (see ANSI/IEEE Std 268-1992).	
*candela	cd		SI unit of luminous intensity.	<i>I</i>
candela per square inch	cd/in ²		Use of the SI unit cd/m ² is preferred.	
*candela per square meter	cd/m ²	nit	SI unit of luminance.	<i>L</i>
candle	cd		The unit of luminous intensity has been given the name <i>candela</i> . Use of the name <i>candle</i> for this unit is deprecated.	
*centi	c (prefix)		SI prefix for 10 ⁻² .	
*centimeter	cm			
centipoise	cP		cP Δ mPa · s. The name centipoise is deprecated (see ANSI/IEEE Std 268-1992).	
centistokes	cSt		cSt Δ mm ² /s. The name centistokes is deprecated (see ANSI/IEEE Std 268-1992).	
*circular mil	cmil		cmil Δ ($\pi/4$) · 10 ⁻⁶ in ² .	
*coulomb	C	c	SI unit of electric charge.	Q
*cubic centimeter	cm ³	cc	Volume. (Preferred SI unit multiple.)	
cubic foot	ft ³			
cubic foot per minute	ft ³ /min	cfm		
cubic foot per second	ft ³ /s			
cubic inch	in ³			
*cubic meter	m ³			
*cubic meter per second	m ³ /s			
cubic yard	yd ³			

curie	Ci	C	Ci Δ 3.7×10^{10} Bq. A unit of activity of a radionuclide. Use of the SI unit, the becquerel, is preferred.	
cycle per second	Hz	c/s, cps, c/sec, cycle	See hertz.	
darcy	D		D Δ cP·(cm/s)·(cm/atm) = 0.986923 μm^2 . A unit of permeability of a porous medium. By traditional definition, a permeability of one darcy will permit a flow of 1 cm ³ /s of fluid of 1 cP viscosity through an area of 1 cm ² under a pressure gradient of 1 atm/cm. Deprecated (see ANSI/IEEE Std 268-1992).	
day	d		day Δ 24 h.	
deci	d (prefix)		SI prefix for 10 ⁻¹ .	
decibel	dB	db, DB		
degree (plane angle)	...°	deg		
degree (temperature)				
degree Celsius	°C	degree centigrade	SI unit of Celsius temperature. The degree Celsius is a special name for the kelvin, used in expressing Celsius temperatures or temperature intervals.	<i>t</i>
degree Fahrenheit	°F		Note that the symbols for °C, °F, and °R are comprised of two elements, written with no space between the ° and the letter that follows. The two elements that make the complete symbol are not to be separated.	
degree kelvin	K		See kelvin.	
degree Rankine	°R			
deka	da		SI prefix for 10.	
dyne	dyn	dyne	dyn Δ 10 ⁻⁵ N. Deprecated (see ANSI/IEEE Std 268-1992).	<i>F</i>
*electronvolt	eV	ev		
erg	erg		erg Δ 10 ⁻⁷ J. Deprecated (see ANSI/IEEE Std 268-1992).	
exa	E		SI prefix for 10 ¹⁸ .	
exbi	Ei		Prefix for 2 ⁶⁰ .	
*farad	F	f, fd	SI unit of capacitance.	<i>C</i>
*femto	f		SI prefix for 10 ⁻¹⁵ .	
femtometer	fm			
foot	ft		ft Δ 0.3048 m.	
foot of water	ftH ₂ O		ftH ₂ O = 2989.1 Pa. (ISO). ¹	
foot per minute	ft/min	fpm		
foot per second	ft/s	fps, ft/sec		
foot per second squared	ft/s ²			
foot pound-force	ft · lbf			
footcandle	fc		fc Δ lm/ft ² . The name <i>lumen per square foot</i> is also used for this unit. Use of the SI unit of illuminance, the lux (lumen) per square meter, is preferred.	

footlambert	fL		fL $\Delta (1/\pi)$ cd/ft ² . A unit of luminance. One lumen per square foot leaves a surface whose luminance is one footlambert in all directions within a hemisphere. Use of the SI unit, the candela per square meter, is preferred.	
gal	Gal		Gal Δ cm/s. Deprecated (see ANSI/IEEE Std 268-1992).	
gallon	gal		1 gal _{UK} = 4.5461 L. 1 gal _{US} Δ 231 in ³ = 3.7854 L.	
gauss	G		The gauss is the electromagnetic CGS unit of magnetic flux density. Deprecated (see ANSI/IEEE Std. 268-1992).	<i>B</i>
gibi	Gi		Prefix for 2 ³⁰ .	
*giga	G	kM	SI prefix for 10 ⁹ .	
gigabyte	GB		GB Δ 10 ⁹ B.	
*gigaelectronvolt	GeV	bev, BeV		
*gigahertz	GHz	kMHz, KMC, Gc/s		
			¹ The term “(ISO)” means that the definition is from ISO 31.	
gilbert	Gb		The gilbert is the electromagnetic CGS unit of magnetomotive force. Deprecated (see ANSI/IEEE Std 268-1992).	
grain	gr		gr Δ lb/7000.	
*gram	g	gm		<i>m</i>
gram per cubic centimeter	g/cm ³			
*gray	Gy		SI unit of absorbed dose in the field of radiation dosimetry.	
*hecto	h		SI prefix for 10 ² .	
*henry	H	Hy, hy	SI unit of inductance.	<i>L</i> <i>P</i> , <i>P_m</i>
*hertz	Hz	cps, c/s, cycle	SI unit of frequency.	<i>f</i> , <i>v</i> <i>B</i>
horsepower	hp		hp Δ 550 ft · lbf/s = 746 W. The horsepower is an anachronism in science and technology. Use of the SI unit of power, the watt, is preferred.	
*hour	h	hr		
inch	in	in.	in Δ 2.54 cm.	
inch of mercury	inHg		inHg = 3386.4 Pa (ISO).	
inch of water	inH ₂ O		inH ₂ O = 249.09 Pa (ISO).	
inch per second	in/s	ips		
*joule	J		SI unit of energy, work, and quantity of heat.	<i>E</i> <i>W</i> <i>Q</i>
*joule per kelvin	J/K		SI unit of heat capacity and of entropy.	<i>S</i>
kelvin	K		In 1967, the CPGM gave the name <i>kelvin</i> to the SI unit of temperature, which had formerly been called <i>degree kelvin</i> , and assigned it the symbol K (without the symbol °).	
kibi	Ki		Prefix for 2 ¹⁰ .	

*kilo	k		SI prefix for 10^3 . The symbol k shall not be used for kilo. The prefix kilo shall not be used to mean 2^{10} (that is, 1024).	
*kilobit per second	kb/s			
*kilobyte	kB		kB $\triangleq$ 1000 bytes.	
kilogauss	kG		Deprecated (see ANSI/IEEE Std 268-1992).	
*kilogram	kg		SI unit of mass.	
kilogram-force	kgf		Deprecated (see ANSI/IEEE Std 268-1992). In some countries the name kilopond (kp) has been used for this unit.	
*kilohertz	kHz			
*kilohm	k Ω			<i>R</i>
*kilometer	km			
*kilometer per hour	km/h			
kilopound-force	klbf		Kilopound-force should not be misinterpreted as kilopond (see kilogram-force).	
*kilovar	kvar			<i>Q</i>
*kilovolt	kV			
*kilovoltampere	kVA	KVA, kva		
*kilowatt	kW			
kilowatthour	kWh		Also kW·h.	
knot	kn		kn $\triangleq$ nmi/h. 0.514 m/s.	
lambert	L		L $\triangleq$ $(1/\pi)$ cd/cm ² . A CGS unit of luminance. One lumen per square centimeter leaves a surface whose luminance is one lambert in all directions within a hemisphere. Deprecated (see ANSI/IEEE Std 268-1992).	
*liter	L		L $\triangleq$ 10 ⁻³ m ³ . In 1979, the CGPM approved L and l as alternative symbols for the liter. Because of frequent confusion with the numeral 1, the letter symbol l is not recommended for U.S. use (see Federal Register notice of December 20, 1990, vol. 55, no. 245, p. 52242). The script <i>l</i> shall not be used as a symbol for liter.	<i>V, v</i>
liter per second	L/s			
*lumen	lm		SI unit of luminous flux.	Φ
lumen per square foot	lm/ft ²		A unit of illuminance and also a unit of luminous exitance. Use of the SI unit, lumen per square meter, is preferred.	
*lumen per square meter	lm/m ²		SI unit of luminous exitance.	<i>M</i>
*lumen per watt	lm/W		SI unit of luminous efficacy.	$K(\lambda)$ K, K_i
*lumen second	lm·s		SI unit of quantity of light.	<i>Q</i>
*lux	lx		1x/lm $\triangleq$ /m ² . SI unit of illuminance.	<i>E</i>
maxwell	Mx		The maxwell is the electromagnetic CGS unit of magnetic flux. Deprecated (see ANSI/IEEE Std 268-1992).	
mebi	Mi		Prefix for 2 ²⁰ .	
*mega	M		SI prefix for 10 ⁶ . The prefix mega shall not be used to mean 2 ²⁰ (that is, 1 048 576).	

megabit per second	Mb/s			
*megabyte	MB		MB Δ 1 000 000 bytes.	
*megaelectronvolt	MeV			
*megahertz	MHz			
*megohm	M Ω	M		
*meter	m		SI unit of length.	<i>L</i>
metric ton	t		t Δ 1000 kg. Use of the name <i>tonne</i> is deprecated in the U.S. (see ANSI/IEEE Std 268-1992).	
mho	S		Ω^{-1} . The name <i>mho</i> was formerly given to the reciprocal ohm. Deprecated; see siemens (S).	
*micro	μ		SI prefix for 10^{-6} .	
*microampere	μ A			
*microfarad	μ F			
*microgram	μ g			
*microhenry	μ H			
microinch	μ in			
*microliter	μ L		See note for liter.	
*micrometer	μ m	μ		
micron	μ m	μ	The name micron is deprecated. Use micrometer.	
*microsecond	μ s			
*microwatt	μ W			
mil	mil		mil Δ 0.001 in.	
mile (statute)	mi		mi Δ 5280 ft = 1609 m.	
mile per hour	mi/h	mph	Although use of mph as an abbreviation is common, it should not be used as a symbol.	
*milli	m		SI prefix for 10^{-3} .	
*milliampere	mA			
millibar	mbar		Use of the bar is strongly discouraged in ANSI/IEEE Std 268-1992, except for limited use in meteorology.	
*milligram	mg			
*millihenry	mH			
*milliliter	mL		See liter.	
*millimeter	mm			
millimeter of mercury	mmHg		mmHg = 133.322 Pa. Deprecated (see ANSI/IEEE Std 268-1992).	
millimicron	nm		Use of the name millimicron for the nanometer is deprecated.	
*millipascal second	mPa · s		SI unit-multiple of dynamic viscosity.	
*millisecond	ms			
*millivolt	mV			
*milliwatt	mW			
*minute (plane angle)	'			
*minute (time)	min		Time may also be designated by means of superscripts as in the following example: 9 ^h 46 ^m 30 ^s .	

*mole	mol		SI unit of amount of substance. The mole is the amount of substance of a system that contains as many elementary entities as there are atoms in 0.012 kg of carbon 12. When the mole is used, the elementary entities shall be specified and may be atoms, molecules, ions, electrons, other particles, or specified groups of such particles.	
month	mo			
*nano	n		SI prefix for 10^{-9} .	
*nanoampere	nA			
*nanofarad	nF			
*nanometer	nm			
*nanosecond	ns			
nautical mile	nmi		nmi $\triangleq$ 1852 m.	
*neper	Np			
*newton	N		SI unit of force.	
*newton meter	N · m			
*newton per square meter	N/m ²		SI unit of pressure or stress. See pascal.	
oersted	Oe	oe	The oersted is the electromagnetic CGS unit of magnetic field strength. Deprecated (see ANSI/IEEE Std 268-1992).	
*ohm	Ω		SI unit of resistance.	
ounce (avoirdupois)	oz		oz $\triangleq$ 1/16 lb = 28.350 g.	
*pascal	Pa		Pa $\triangleq$ N/m ² . SI unit of pressure or stress.	
*pascal second	Pa · s		SI unit of dynamic viscosity.	
pebi	Pi		Prefix for 2^{50} .	
*peta	P		SI prefix for 10^{15} .	
phot	ph		ph $\triangleq$ lm/cm ² . CGS unit of illuminance. Deprecated (see ANSI/IEEE Std 268-1992).	
*pico	p		SI prefix for 10^{-12} .	
*picofarad	pF			
*picowatt	pW			
pint	pt		pt (U.K.) = 0.568 26 L. pt (U.S. dry) = 0.550 6 L. pt (U.S. liquid) = 0.473 18 L.	
poise	P		Deprecated (see ANSI/IEEE Std 268-1992).	
pound (avoirdupois)	lb		lb $\triangleq$ 0.453 592 37 kg.	
pound per cubic foot	lb/ft ³			
pound-force	lbf		lbf = 4.4482 N.	
pound-force foot	lbf · ft			
pound-force per square foot	lbf/ft ²			
pound-force per square inch	lbf/in ²	psi	Although use of the abbreviation psi is common, it should not be used as a symbol.	
poundal	pdl		pdl $\triangleq$ lb · ft/s ² = 0.1383 N	
quart	qt		qt (U.K.) = 1.1365 L. qt (U.S. dry) = 1.1012 L. qt (U.S. liquid) = 0.946 35 L.	
rad	rd		rd $\triangleq$ 0.01 Gy. A unit of absorbed dose in the field of radiation dosimetry. Use of the SI unit, the gray, is preferred.	

*radian	rad		SI unit of plane angle.	
rem	rem		rem Δ 0.01 Sv. A unit of dose equivalent in the field of radiation dosimetry. Use of the SI unit, the sievert, is preferred. 1 rem = 0.01 Sv.	
revolution per minute	r/min		Although use of rpm as an abbreviation is common, it should not be used as a symbol.	
revolution per second	r/s			
roentgen	R		A unit of exposure in the field of radiation dosimetry.	
*second (plane angle)	"		1" = $4.848 \cdot 10^{-6}$ rad.	
*second (time)	s		SI unit of time.	
*siemens	S		S Δ Ω^{-1} . SI unit of conductance.	
*sievert	Sv		SI unit of dose equivalent in the field of radiation dosimetry.	
slug	slug		slug Δ lbf $\cdot$ s ² /ft = 14.594 kg.	
square foot	ft ²			
square inch	in ²			
*square meter	m ²			
*square meter per second	m ² /s		SI unit of kinematic viscosity.	
*square millimeter per second	mm ² /s		SI unit-multiple of kinematic viscosity.	
square yard	yd ²			
*steradian	sr		SI unit of solid angle.	
stilb	sb		sb Δ cd/cm ² . A CGS unit of luminance. Deprecated (see ANSI/IEEE Std 268-1992).	
stokes	St		Deprecated (see ANSI/IEEE Std 268-1992).	
tebi	Ti		Prefix for 2 ⁴⁰ .	
*tera	T		SI prefix for 10 ¹² .	
terabyte	TB		TB Δ 10 ¹² B.	
*tesla	T		T Δ N/(A $\cdot$ m) ² Δ Wb/m ² . SI unit of magnetic flux density (magnetic induction).	
therm	thm		thm Δ 100 000 Btu.	
ton (short)	ton		ton Δ 2000 lb.	
ton, metric	T		t Δ 1000 kg. Use of the <i>tonne</i> for this unit is deprecated in the U.S. (see ANSI/IEEE Std 268-1992).	
torr	torr		A unit of pressure equal to 0.001316 atmosphere; named after Torricelli.	
*(unified) atomic mass unit	u		The (unified) atomic mass unit is defined as one-twelfth of the mass of an atom of the carbon- 12 nuclide. Use of the old atomic mass unit (amu), defined by reference to oxygen, is deprecated.	
*var	var		IEC name and symbol for SI unit of reactive power.	
*volt	V		SI unit of voltage.	
*volt per meter	V/m		SI unit of electric field strength.	
*voltampere	VA	va	IEC name and symbol for SI unit of apparent power.	
*watt	W		SI unit of power.	
*watt per meter kelvin	W/(m $\cdot$ K)		SI unit of thermal conductivity.	

*watt per steradian	W/sr		SI unit of radiant intensity.
*watt per steradian square meter	(W/sr · m ²)		SI unit of radiance.
watthour	Wh		
*weber	Wb		Wb $\triangleq$ V·s. SI unit of magnetic flux.
yard	yd		yd $\triangleq$ 0.9144 m.
year	a		Also W·h.
yobi	Yi		Prefix for 2 ⁸⁰ .
yocto	y		SI prefix for 10 ⁻²⁴ .
yotta	Y		SI prefix for 10 ²⁴ .
zebi	Zi		Prefix for 2 ⁷⁰ .
zepto	z		SI prefix for 10 ⁻²¹ .
zetta	Z		SI prefix for 10 ²¹ .

D. Miscellaneous Alphabetical Abbreviations, Acronyms, and Symbols

NOTE: Key: fn—function name (roman); s—symbol (italic); u—unit abbreviation (roman);

*—acronyms that must be defined in text.

A	
<i>A</i>	(s) Hermitian conjugate of <i>A</i>
Å	(u) angstrom
ab	(prefix) denotes absolute system of (CGS) units. Abampere, abcoulomb, abvolt, abohm, abfarad, abmho, abhenry (use not recommended, see units list)
abs	absolute
ABS	air-bearing surface
Ac	alternating current
ACB	air circuit breaker*
ACSR	steel-reinforced aluminum cable*
AD	attention display*
A–D, A/D	analog-to-digital
ADF	automatic direction finder*
a.e.	almost everywhere (in equations)
AEW	airborne early warning*
AF	audio frequency*
AFB	Air Force Base
AFC	automatic frequency control*
AFM	atomic force microscopy
AGC	automatic gain control*
AGFM	alternating gradient force magnetometer
AGM	arithmetical–geometric mean*
A·h (u)	ampere hour
Ai (fn)	Airy integral
AM	amplitude modulation

A.M.	ante meridiem (morning)
ama	automatic message accounting*
AND	(small caps) logical AND operation
ANI	automatic number identification
ANN	artificial neural network*
antilog (fn)	antilogarithm
AOGM	accelerated optimum gradient method*
AOPT	air-operated press type*
APD	avalanche photodiode
API	air position indicator*
AQL	acceptable quality level
AR	antireflection*; autoregressive*
arcsin arccos arctan arccot arcsec arccsc	(fn) inverse trigonometric functions
arg	(fn) argument
ARMA	autoregressive moving average*
a.s.	almost surely (in equations)
ASE	amplified spontaneous emission*
ASIC	application specified integrated circuit*
ASK	amplitude-shift keying
ASW	antisubmarine warfare* (note: for acoustic surface wave use SAW)
at (u)	technical atmosphere: 1 kgf/cm
At (u)	ampere turn (note: no longer in use; change to A)
ATM	asynchronous transfer mode*
atm (u)	atmosphere
ATR	antitransmit receive*
ATT	avalanche transit time*
av	average (subscript)
AVC	automatic volume control*
avg (fn)	average (use av as subscript)
AWE	asymptotic wave evaluation*
AWG	American wire gauge
AWGN	additive white Gaussian noise*
B	
bar (u)	bar
barye (u)	barye: microbar (use not recommended; see units list)
bbl (u)	barrel (see units list)
bcc	body-centered cubic (of crystals)
BCD	binary coded decimal
BCH	Bose–Chaudhuri–Hocquenghen (codes)

BCT	bushing current transformer*
Bd (u)	baud* (see units list)
B–E	base–emitter source
Be	Baume
bei, ber (fn)	Kelvin forms of Bessel function
BEM	boundary-element method
BER	bit error rate*
BeV, bev (u)	use GeV
BFO	beat-frequency oscillator*
B–H B–H curve:	curve of magnetic induction (magnetic flux–density) versus magnetic intensity (field intensity) B–H relationship. B–H loop: hysteresis loop
Bhp	brake horsepower*
Bi (fn)	Airy integral: (u) bit: = 10 A*
BIL	basic impulse insulation level*
BJT	bipolar junction transistor*
BMEP	brake mean effective pressure*
bpi (u)	bit per inch: use b/in
bps (u)	bit per second: use b/s
BPSK	binary phase-shift keying
BRA	biased rectifier amplifier*
BS	breaking strength*
BS	British Standards*
B&S	Brown and Sharpe gauge*
BSF	bulk shielding facility*
BSL	basic switching surge insulation level*
BTU	(u) British thermal unit
BWG	Birmingham wire gauge*
BWK	Brillouin–Wentzel–Kramers (method)*
BWO	backward-wave oscillator*
BWR	boiling water reactor*
C	
C (u)	coulomb
°C (o)	degree Celsius
c (u)	cycle: use Hz; centi- (prefix to unit abbreviation)
<i>c</i> (s)	speed of light in a vacuum
cal (u)	calorie (use not recommended; see units list)
CATV	community antenna television system
cc (u)	cubic centimeter: use cm ³
c.c.	complex conjugate (in equations)
CCB	coin collecting box (British telephones)*
CCD	charge-coupled device*
CCR	closed-cycle refrigerator*
cd (u)	candela

cdf	cumulative distribution function*
CDMA	code division multiple access*
CDO	community dial offices*
CD-ROM	compact disk read-only memory
cdrx	external critical damping resistance: use caps*
CEMF	counterelectromotive force*
cf.	compare
cfm (u)	cubic feet per minute: use ft ³ /min
cfs (u)	cubic feet per second: use ft ³ /s
CGS	centimeter-gram-second (system of units)
Ci (fn)	cosine integral; (u) curie
CIM	computer integrated manufacturing*
CIR	carrier-to-interference ratio*
ckVA	capacitive kilovoltamperes (write out)
cmil (u)	circular mil
CMOS	complementary metal–oxide–semiconductor
CNN	cellular neural network
COP	coefficient of performance*
cos	(fn) cosine
cosec	(fn) cosecant: use csc
cosh	(fn) hyperbolic cosine
cot	(fn) cotangent
coth	(fn) hyperbolic cotangent
covers	(fn) coversine
cP (o)	centipoise (see units list)
CPFSK	continuous phase frequency-shift keying*
CPM	continuous phase modulation*
CPSK	continuous phase-shift keying; coherent phase-shift keying*
CPU	central processing unit
CRO	cathode-ray oscilloscope
CRS	cold-rolled steel*
CRT	cathode-ray tube
c/s (u)	cycle per second: use Hz
csc (fn)	cosecant
csch (fn)	hyperbolic cosecant cs (u) centistokes: use cSt or write out (see units list)
CSP	completely self-protected
cSt (u)	centistokes (see units list)
CSV	corona-starting voltage
CT	current transformer*
CTC	centralized traffic control
ctn (fn)	cotangent: use cot
curl (fn)	curl
CV	capacitance–voltage
CVD	chemical vapor deposited
CW	continuous wave*

D	
DA	design automation
dB (u)	decibel
dc	direct current (DC at start of sentence or in article title)
DC	directional coupler
DDA	digital differential analyzer*
DDD	direct distance dialing*
DE	disruptive effect*
det (fn)	determinant
DF	direction finder*; deuterium fluoride; degree of freedom*
DFB	distributed feedback
DFT	discrete Fourier transform*
diag	(diagonal)
diam	diameter
DIC	Diploma of membership in Imperial College of Science and Technology
div (fn)	divergence; division (u) in charts
DMA	direct memory access*
DME	distance-measuring equipment*
DOD	diameter over dielectric; Department of Defense
DOF	degree of freedom (unit)
DP	dial pulse*
DPCM	differential pulse code modulation*
DPDT	double-pole double-throw switch*
DPH	diamond pool hardness*
DPQSK	differential quadrature phase-shift keying*
DPSK	differential phase-shift keying*
DRCPR	differential reactive current protective relay*
DRO	destructive readout*; doubly resonant oscillator
DS	dielectric strength*; direct sequence*
DSB	double sideband*
DSP	digital signal processor
DVP	differential vapor pressure*
DWT	discrete wavelet transform*
dyn (u)	dyne
E	
EB	emergency bank*
EC	eddy current; electrical conductivity* (grade of Al)
ECG	electrocardiogram
ECL	emitter-coupled logic*
ECM	electronic countermeasures
ECT	eddy current testing
ED	enforced draft

EDFA	erbium-doped fiber amplifiers*
EDP	electronic data processing
EDS	energy dispersive spectrometer
EDX	energy dispersive X-ray
EEG	electroencephalogram
EHD	electrohydrodynamic*
EHF	extremely high frequency*
EHPS	extra heavy iron pipe size*
EHV	extra high voltage
Ei (fn)	exponential integral
ELF	extremely low frequency*
EM	electromagnetic*

EMF	electromotive force*
EMI	electromagnetic interference*
ems	expected value of mean square*
EMU	electromagnetic units
EOF	end of file
erf (fn)	error function
erfc (fn)	complementary error function
erg (u)	erg
ERP	effective radiated power*
ESS	electrical sheet steel*
ESU	electrostatic units
eV (u)	electronvolt
EXOR	EXCLUSIVE-OR circuit (small caps)
exp (fn)	exponential function
exsec (fn)	exsecant
F	
f (f-stop, f/22)	ratio of focal length to aperture
F (u)	farad
°F (u)	degree Fahrenheit
FA	forced-air-cooled transformer*
fcc	face-centered cubic (of crystals)
FCC	Federal Communications Commission
FD	flux density*
FDA	finite difference approximations*
FDM	frequency-division multiplexing*
FDMA	frequency-division multiple access*
FDTD	finite-difference time domain*
FEA	finite-element analysis
FET	field-effect transistor
ff.	following pages
FFT	fast Fourier transform*
FIFO	first-in first-out
FIM	field intensity meter*
FIR	finite-impulse response*
fL (u)	footlambert
FL	full load
FM	frequency modulation
FMFB	FM feedback receiver*
FMR	frequency of maximum reliability*; ferromagnetic resonance
FPGA	field-programmable gate array*
fpm, fps (u)	feet per minute: use ft/min; feet per second: use ft/s
FS	full scale
FSK	frequency-shift keying*

FSM	finite-state machine*
ft (u)	foot
FTL	flat tie-line*
FTP	file transfer protocol
FW	full wave
FWHM	full-width at half-maximum*
FWM	four-wave mixing*
<u>G</u>	
G	giga- (prefix to unit abbreviations) = 10^9
G (u)	gauss
g	acceleration of gravity, “gee force”; use as unit with metric prefix, as in 3 mg
G (s)	gravitational constant
Gal (u)	gal (gravitational unit)
gal (u)	gallon
Gb (u)	gilbert
GCA	ground-controlled approach*
gcd	greatest common denominator (may be function name)
GenAI	generative artificial intelligence
GLB	greatest lower bound*
GMD	geometric mean distance*
GMEC	generalized minimum effort control*
GMF	geometric mean frequency
GMR	geometric mean radius
GMT	Greenwich mean time
gpd (u)	gallon per day: use gal/day
GPS	Global Positioning System
GPU	graphical processing unit, General Public Utilities*
grad (fn)	gradient
GSE	ground support equipment*
GTD	geometrical theory of diffraction
GUI	graphical user interface
GW	ground wire
<u>H</u>	
h (s)	Planck’s constant
H (u)	henry
H (s)	magnetic intensity; magnetic field strength
hav, havers (fn)	haversine
HBT	heterojunction bipolar transistor
hcp	hexagonal close-packed (of crystals)
HD	hard-drawn*
HDBC	hard-drawn bare copper*
HDC	hard-drawn copper*

HDD	hard disk drive
HDT	hard-drawn tubing*
HEMT	high-electron mobility transistor
HF	high frequency; hydrogen fluoride
HFET	heterojunction FET
HG	mercury
hipot	high potential (write out)
hp (u)	horsepower
HTC	high-tension cable*
HTML	hypertext markup language
HV	high voltage
HVdc	high voltage direct current
Hz (u)	hertz
I	
I (s) current (fn)	imaginary part of: use Im
IACS	International Annealed Copper Standard*
IC	impedance compensation*; integrated circuit
ICW	interrupted continuous wave*
ID	inside diameter; induced draft*; interdigital*
IDP	integrated data processing*
IF	intermediate frequency
iff	if and only if
IFT	interfacial tension*
IGFET	insulated-gate field-effect transistor
i.i.d.	independent identically distributed*
IIR	infinite-impulse response
ILS	instrument landing system*
Im (fn)	imaginary part of
IM	intermediate modulation
IMPATT	impact ionization avalanche transit time (diode)
INE	irredundant normal equivalent*
inf (fn)	infimum
int (fn)	integer value of
I/O, I–O	input–output
IoT	Internet of Things*
IP	Internet Protocol
ips (u)	inch per second: use in/s
IPS	iron pipe size; international pipe standard*
IR	infrared
I/R	current–resistance
ISB	independent sideband*
ISE	integral of squared error*
ISI	intersymbol interference

itae	integral of time-multiplied absolute value of error
ITI	inter-track interference
I – V (s)	current–voltage (characteristic or curve)
IVA	induced voltamperes
IX	current–reactance (drop)
IZ	current–impedance
J	
J (u)	joule
JFET	junction field-effect transistor
JPEG	Joint Photographers Expert Group
K	
k	kilo (prefix to unit abbreviations) = 10^3
K (u)	Kelvin
Kayser (u)	= cm^{-1} (wavenumber)
kbits (u)	kilobits per second: use kb/s
KCL	Kirchhoff's current law
kcm, KCM (u)	thousand circular mils: use kcmil
kg (u)	kilogram
KGO, KGOe, KGoe, KgOe (u)	use kO·Oe
kgp (u)	kilogrampois (French): use kg
kG.Oe (u)	kilogauss oersted
kip	thousand pounds
kn (u)	knot (nautical mile per hour)
KOH	potassium hydroxide
kp (u)	kilopound (German): use kg
kt (s)	Boltzmann's constant $\times$ time
KVL	Kirchhoff's voltage law
kVp (u)	kilovolt peak*
L	
l (u)	liter
L (u)	lambert
LAN	local area network
lb (u)	pound
lbf (u)	pound-force
LC	inductance–capacitance
lcm	least common multiple (may be function name)
LCR	inductance–capacitance–resistance
LCS	load current substation*
LDC	line drop compensator*; load division circulation
LED	light-emitting diode

LF	low-frequency
LHP	left-half plane*
LHS	left-hand side*
Li (fn)	logarithmic integral
lim (fn)	limit
l.i.m. (fn)	limit in the mean
L–L	line to line*
lm (u)	lumen
LMLT	locus of major loop tips*
LMS	least mean square
LMT	local mean time*
ln (fn)	natural logarithm (base e)
L–N	line to neutral*
LNA	low noise amplifier
LO	local oscillator*
log, $\log_n$ (fn)	logarithm, logarithm base n (where $n = 2, 10$, etc.)
LP	linear programming*
LPE	liquid phase epitaxy*
LR	inductance–resistance
LRC	load ratio control*
LSB	least significant bit
LSI	large-scale integration*; large-scale integrated*
LST	local standard time
LTC	load tap-changing*
LTE	long-term evolution
LTS	laser-triggered switching*
LUF	lowest usable frequency*
lx (u)	lux
M	
m (u)	meter; milli- (prefix to unit abbreviations) = 10^{-3}
M	mega- (prefix to unit abbreviations) = 10^6 ; mole
MAG	maximum available gain
MAP	maximum <i>a posteriori</i>
max (fn)	maximum; also used as subscript
MC	Monte Carlo
mcm, MCM (u)	thousand circular mils: use kemil
mc/mM (u)	millicuries per millimole: use mCi/mM
MCS	multicircuit substation*
MCT	movable core transformer*
MCW	modulated continuous wave*
MDF	manual direction finder*
MDS	minimum detectable signal
MEMS	micro-electromechanical systems

MESFET	metal–semiconductor field-effect transistor
MEW	microwave early warning*
MF	medium frequency*
MFM	magnetic force microscopy
MFSK	minimum frequency-shift keying
MGO (u)	megagauss oersted: use MG·Oe
MG·Oe (u)	megagauss oersted
MHD	magnetohydrodynamics
mho (u)	mho (also Ω^{-1})
mi (u)	mile
MIM	metal–insulator–metal
MIMO	multi-in multi-out*
mio (fn)	minimum; also used as subscript
MIS	metal–insulator–semiconductor*
MKS	meter–kilogram–second (system of units)
ml	milliliter
MLE	maximum-likelihood estimation*
MLSD	maximum-likelihood sequence detector
MLSE	maximum-likelihood sequence estimator*
MMF	magnetomotive force
mmHg (u)	millimeter of mercury
MMIC	monolithic microwave integrated circuit*
mm ₂ O (u)	millimeter of water
mmse	minimum mean square error
MOCVD	metal–organic chemical vapor deposition*
mod	modulo
MOKE	magneto optic Kerr effect
MoM	method of moments*
MOS	metal–oxide–semiconductor
MOSFET	MOS field-effect transistor
MOST	MOS transistor
MOVPE	metal–organic vapor phase epitaxy*
MPEG	Moving Pictures Expert Group
MPIE	mixed potential integral equation
MRAM	magnetic random access memory
MRI	magnetic resonance imaging
MSB	most significant bit
mse	mean square error
MSIC	medium scale integrated circuits*
MTBE	mean time between explosions
MTBF	mean time between failures*
MTI	multiple target indicator*; moving target indicator
MTJ	magnetic tunnel junction
MTL	multiconductor transmission line

MU	multiple unit*
MUF	maximum usable frequency*
MVQE	minimum variance quantum estimator
Mx (u)	maxwell
MZI	Mach–Zehnder interferometric*
<u>N</u>	
<i>n</i>	nano (prefix to unit abbreviations) = 10 ⁻⁹
N (u)	newton
NA	numerical aperture*
NAND	NOT-AND circuit (small caps)
nat (u)	nat
NC	diode negative-conductance diode*
NDRO	nondestructive readout
NDT	nondestructive testing*
NIC	negative impedance converter*
NIR	near infrared response*
nit (u)	nit
Nkw-hr (u)	net kilowatthour: use net kW·h
NL	no load
nmi (u)	nautical mile
NMR	nuclear magnetic resonance*
NOR	NOT-OR circuit (small caps)
NP	nameplate (rating)
Np (u)	neper
n-p-n	semiconductor forms: Roman, lowercase, hyphens
NRZ	nonreturn to zero*
NTC	negative temperature coefficient*
NWP	network protector
<u>O</u>	
OCB	oil circuit breaker*
OCR	oil circuit recloser*
OD	outside diameter
Oe (u)	oersted
OEIC	optoelectronic integrated circuit*
OFDA	optical-fiber frequency-domain analysis*
OGM	optimum gradient method
OOK	on–off keying
OOP	object-oriented programming*
opt (fn)	optimum: also used as subscript
OR	OR circuit (small caps)
OSM	omni spectra miniature
OTDM	optical time-division multiplexing*

<i>O</i> -wave	ordinary-wave (ionogram)
oz (u)	ounce
P	
p	pico- (prefix to unit abbreviations) = 10^{-12}
P (u)	poise
Pa (u)	pascal
PAE	power-added efficiency
PAM	pulse-amplitude modulation*
PAX	private automatic exchange*
PBX	private branch exchange*
pc (u)	parsec
PC	personal computer
PCM	pulse-code modulation*; pulse-count modulation*
PD	potential difference*
pdf	probability density function*
pdl (u)	poundal (see units list)
PDM	pulse-duration modulation*
P_e (s)	probability of error
PER	probability of error
PES	position error signal
PF	power factor*
ph (fn)	phase
pH	power of hydrogen (acidity or alkalinity of solution)
PI	polarization index
PID	proportional–integral–differential*
PILC	paper-insulated lead-covered*
PIN	use p-i-n for diodes, etc.
p-i-n	semiconductor forms: Roman, lowercase, hyphens
PL/I	a programming language
PLC	power line carrier*
PLL	phase-locked loop*
PM	phase modulation*
P.M.	post meridiem (small caps)
PML	perfectly matched layer
PMMA	polymethyl methacrylate*
PMR	perpendicular magnetic recording
p-n-i-p	semiconductor forms: Roman, lowercase, hyphens
p-n-p	semiconductor forms: Roman, lowercase, hyphens
POD	para-operational device*
POW _p (u)	picowatts psophometrically weighted at a point of zero relative level*
pp, p-p	peak to peak*
PPI	plan-position indicator*
ppm (u)	parts per million; pulse per minute*

PPM	pulse-position modulation*
pps (u)	pulse per second*
Pr (fn)	probability (appears as $\Pr x \mid x = U$)
PRA	pulse relaxation amplifier
PRF	pulse-repetition frequency*
PRML	partial response maximum likelihood
Prob.,	$P, x \cdot$ use Pr (usually)
PRR	pulse-repetition rate*
PSD	power spectral density
PSF	power separation filter*
psi (u)	pounds per square inch: change to lb/in ² unless paper also contains psia and/or psig
psia (u)	pound-force per square inch absolute (stet)
psig (u)	pound-force per square inch gauge (stet)
PSK	phase-shift keying*
PTM	pulse-time modulation
p.u.	per unit
PVC	polyvinyl chloride*
PWL	piecewise linear
PWM	pulse width modulation*
PWR	pressurized water reactor*
PZT	lead zirconate titanate
Q	
Q	quality factor; figure of merit
QAM	quadrature-amplitude modulation*
Q.E.D.	quod erat demonstrandum (end of proof) (set flush right)
QoS	quality of service
QP	quasi-peak*
QPSK	quaternary phase-shift keying
QW	quantum well*
R	
R (u)	roentgen
R (fn)	real part of: use Re
°R (u)	degrees Rankine
rad (u)	radian
RAM	random access memory
RB	circuit transient blocking relay circuit*
RC	resistance-capacitance
RCF	radar cross section*
R&D	research and development
Re (fn)	real part of: use Re (be sure of this meaning before changing)
redox	reduction-oxidation

rem (u)	Roentgen equivalent, man
RF	radio frequency
RFI	radio frequency interference*
RFU	reclosing fuses*
RH	relative humidity*
RHS	right-hand side*
RI	radio interference*
RIFI	radio interference and field intensity*
RIL	radio interference level*
RIN	relative intensity noise*
<i>RL</i>	resistance–inductance
RMI	radiomagnetic indicator*
rms	root-mean-square (error); root mean square
ROM	read-only memory
rpm (u)	revolution per minute: use r/min
rps (u)	revolution per second: use r/s
RSG	recurrent surge generator*
RTD	resistance temperature detectors
RV	random variable
<i>RX</i>	resistance–reactance
<u>S</u>	
s (u)	second
S (u)	siemens
SAR	specific absorption rate
SATT	Strowger Automatic Toll Ticket*
SAW	surface acoustic wave*
SC	switched-capacitor*(adj)
SCA	steel-reinforced aluminum cable*
SCC	signal component control*
scfm	standard cubic feet per minute*
SCL	space-charge limited*
scr	short-circuit ratio*
SCR	silicon-controlled rectifier
sec (fn)	secant; (u) second: use s; second of arc*
sech (fn)	hyperbolic secant
SEM	scanning electron microscope
SF	single frequency*
SGML	standard generalized markup language
sgn (fn)	signum function
SHF s	upper high frequency*
SI	severity index*; Systeme International d’Unites (International System of Units)
Si, si (fn)	sine integral
sin (fn)	sine

sinc (fn)	$\text{sinc } x = (\sin x) / x$
sinh (fn)	hyperbolic sine
SINR	signal-to-interference-plus-noise ratio*
SIR	signal-to-interference ratio
SISO	single-in, single-out*
SLAR	side looking airborne radar
SLG	single line to ground
SMSA	standard metropolitan statistical area
S/N	signal-to-noise ratio
SNR	signal-to-noise ratio
SoC	system-on-chip*
SPDT	single-pole double-throw (switch)*
SPICE	Simulation Program with Integrated Circuit Emphasis
SPT	single-pole type
sq square:	if on a unit, change to ²
SQUID	superconducting quantum interference device
sr (u)	steradian
SR	saturable reactor*
SS	subsystems*
SSB	single sideband*
s.t.	subject to
St (u)	stokes
sterad (u)	steradian: use sr
SUL	soft underlayer
SUMT	sequential unconstrained minimization techniques
sup (fn)	supremum
sus	Saybolt universal seconds (oil viscosity)*
sw	sine wave*
SW	short wave*
SWG	standard wire gauge*
SWR	standing-wave ratio*
T	
t (u)	tonne
T (u)	tesla
tan (fn)	tangent
tanh (fn) hyperbolic tangent	
TCUL	tap-changing under load*
TDM	time-division modulation*; time-division multiplexing*
TDMA	time-division multiple access*
TE	transverse electric (appears as TE_{01}^0 and TE_{01})
TEFC	totally enclosed fan-cooled*
Telex	teleprinter exchange*

TEM	transverse electromagnetic
TFT	thin-film transistor*
tg (fn)	tangent: use tan
th (u)	thermie
TIF	telephone influence factor*
TLM	transmission-line matrix
TM	transverse magnetic
tof	thermal ohms per foot (spell out)
torr (u)	torr
tpc (u)	turns per centimeter: turns/cm
TPC	turns per coil*
tr (fn)	trace
Tr	transpose
TSS	time sharing system
TTL	transistor–transistor logic
TTY	teleprinter
tu	traffic units*
TVI	television interference*
TWA	traveling-wave amplifier*
TWM	traveling-wave maser*
TWP	traveling-wave phototube*
TWT	traveling-wave tube
U	
UHF	ultrahigh frequency
ult (fn)	ultimate
UPS	uninterruptible power system*
	uniform <i>RC</i> sections (stet overbar)
URL	uniform resource locator
XRD	X-ray diffraction
UT	universal time
UTS	ultimate tensile strength
UV	ultraviolet
V	
V (u)	volt
<i>V</i> (s)	voltage
VA (u)	voltampere; Viterbi algorithm*
var (u)	var
VCL	varnished-cambric lead-covered*
VCO	voltage-controlled oscillator*
VCW	type V copper weld*
VDS	voltage divider switching*
ver, vers (fn)	versine

VF	voice frequency*
VFO	variable-frequency oscillator*
VHF	very high frequency*
$V-I$	voltage–current (characteristic of curve)
VLF	very low frequency*
VLSI	very large scale integration*
VOR	very high-frequency omnidirectional radio
VR	voltage regulator*
VSF	vestigial sideband*
VSWR	voltage standing-wave ratio
VTB	voltage time to breakdown*
VTVM	vacuum-tube voltmeter
vu	volume units*
<u>W</u>	
W (u)	watt
WAN	wide area network
Wb (u)	weber
WDM	wavelength-division multiplexing*
WDMA	wavelength-division multiple access*
WKB	Wentzel–Kramer–Brillouin*
wpl, w.p.l.	with probability 1*
wrt, w.r.t.	with respect to
WT	watertight*
wt%	weight percent
<u>X</u>	
XPM c	cross-phase modulation
XOR	EXCLUSIVE-OR circuit (small caps)
X-wave	extraordinary-wave (ionogram)
<u>Y</u>	
YAG	yttrium aluminum garnet
yd (u)	yard
YIG	yttrium iron garnet

Factor by Which the Unit Is Modified	Prefix	Symbol
1000000000000 = 10^{12}	tera	T
1000000000 = 10^9	giga	G
1000000 = 10^6	mega	M
1000 = 10^3	kilo	k
100 = 10^2	hecto	h
10 = 10^1	deka	da

$0.1 = 10^{-1}$	deci	d
$0.01 = 10^{-2}$	centi	c
$0.001 = 10^{-3}$	milli	m
$0.000001 = 10^{-6}$	micro	μ
$0.000000001 = 10^{-9}$	nano	n
$0.000000000001 = 10^{-12}$	pico	p
$0.000000000000001 = 10^{-15}$	femto	f
$0.000000000000000001 = 10^{-18}$	atto	a

For prefixes indicating powers of 2, see Table 7 at the [NIST site](#).

E. Inclusive Language Guide

Requirements:

- Use “people-first language,” i.e., the person has X; has been diagnosed with X; uses a/n X; etc.
- Do not spell out the acronyms LGBTQIA+, LGBT, LGBTQ, LGBTQIA.
- Highlighted terms must be avoided at all times.

Insensitive Term/Phrase	Replace With	Definition/Background	Additional Notes
A.D. (when referencing history/time)	C.E., common era	Abbreviation of the Latin phrase anno Domini, translated as “the year of the Lord.” Traditionally, it is used to date years after the birth of Jesus	This contradicts AP style.
Able-bodied	non-disabled/does not have a disability		
Afro-American/Negro/Colored/Nigger (in reference to race)	Avoid in all instances; African American*; Black†	People of African descent have widely varied cultural backgrounds, family histories, and family experiences. Some will be from Caribbean islands, Latin America, various regions in the United States, countries in Africa, or elsewhere. Some American people of African ancestry prefer “Black,” and others prefer “African American”; both terms are acceptable.	* Specific to people of specific African descent; not to be used as an umbrella for people of African ancestry worldwide. † Widely accepted to encompass multiple ethnicities and/or national origins.
B.C. (when referencing history/time)	B.C.E., before common era	Literally, before Christ or the Christian era.	This contradicts AP style.
Black box	Closed box		Preferred term from IEEE Thesaurus

Blacklist; black list	Blocklist; block list	The Hollywood blacklist was instituted by the House Un-American Activities Committee in 1947 to block screenwriters and other Hollywood professionals who were purported to have Communist sympathies from obtaining employment. A list or compilation that identifies entities that are denied, unrecognized, or ostracized. The term's racist connotations derive from the idea that black equates to negative; this view can be controversial.	Preferred term from IEEE Thesaurus
Blind	blind*; limited vision; low vision; partially sighted†	According to the American Foundation for the Blind, the term “legally blind” denotes a person with 20/200 visual acuity or less. Therefore, “blind” or “legally blind” is acceptable for people with almost complete vision loss. Many people with vision loss are not considered blind.	* Use only with people who are “legally blind” (a person with 20/200 visual acuity or less) † Used most often in British publications
Blind channel estimation	Source channel estimation		Preferred term from IEEE Thesaurus
Blind equalizers	Source equalizers		Preferred term from IEEE Thesaurus
Blind signal separation	Mix signal separation		Preferred term from IEEE Thesaurus
Blind source separation	Mix source separation		Preferred term from IEEE Thesaurus
Caretaker	caregiver	A caregiver is an individual who assists another, including a person with a disability, with his or her daily life, according to Merriam-Webster.	Caretaker denotes taking care of property; Caregiver denotes giving care to people.
Caucasian	European American*; White	The use of the term “Caucasian” as an alternative to “White” or “European” is discouraged because it originated as a way of classifying White people as a race to be favorably compared with other races. As with all discussions of race and ethnicity, it is preferable to be more specific about regional (e.g., Southern European, Scandinavian) or national (e.g., Italian, Irish, Swedish, French, Polish) origin when possible.	* Adjust as needed for location (i.e., European, European American, European Australian, etc.)
Chairman/woman	chairperson		

Committed suicide	died by suicide		
Crazy/loony/mad/ psycho/nuts/deranged/ insane/insanity/ mentally deranged/ psychopathology	mental illness*; mental disorder*; psychopathology(ical)	Once commonly used to describe people with mental illness; commonly used informally to denote mental instability or mental illness.	* Except in a quote or when referring to a criminal defense.
Cripple	Avoid in all instances*; use people first language and their diagnosis, i.e., “person with X”	Merriam-Webster defines the noun “cripple” as “a lame or partly disabled person or animal” and as “something flawed or imperfect.” It is also used as a verb. The word dates back to Old English, where it was related to words that meant to creep or bend over.	
Deaf	D(d)eaf*; hard of hearing	Having total or partial hearing loss.	* NCDJ Recommendation: Lowercase when referring to a hearing-loss condition or to a deaf person who prefers lowercase. Capitalize for those who identify as members of the Deaf community or when they capitalize Deaf when describing themselves. “D(d)eaf” should be used as an adjective, not as a noun; it describes a person with profound or complete hearing loss. Other acceptable phrases include “woman who is deaf” or “boy who is hard of hearing.” When quoting or paraphrasing a person who has signed their responses, it’s appropriate on first reference to indicate that the responses were signed. It’s acceptable to use the word “said” in subsequent references. Per the National Association of the Deaf, “D(d)eaf” is acceptable.

Deformed/deformity (when referencing a person)	Avoid in all instances*; refer to specifics description rather than generalized term of deformity	Merriam-Webster defines as a part of the body that does not have the typical or expected shape	
Dumb (mute)	nonverbal	Once widely used to describe a person who could not speak and also implied the inability to express oneself; however, someone who does not use speech still may have the ability of expression.	
Dwarf/vertically challenged/midget	Dwarf*; short stature; little person	Dwarfism is a medical or genetic condition that results in a stature below 4'10," according to Little People of America. The terms "little people" and "little person" refer to people of short stature and have come into common use since the founding of the Little People of America organization in 1957.	* Use only when applied to a medical diagnosis or in a quote.
Gay marriage/same-sex marriage	Marriage*		* When writing about the inability to legally marry, use "exclusion from Marriage" or "denial of marriage."
Handicap	Use people first language; refer to the person's condition	The Oxford English dictionary defines a handicap as "a condition that restricts a person's ability to function physically, mentally or socially."	
Homosexual/Gay	gay*; lesbian, bi(sexual)†	Short form term to reference gay, lesbian, and bisexual orientations, though not transgender people or gender identity. Anti-gay activists use "homosexual" as a slur to stigmatize gay people by reducing their lives to purely sexual terms.	* Only when used as an adjective (i.e., gay people); † Use bisexual as an adj. and as needed on first reference for clarity, otherwise default to bi.
Indian (when referencing Indigenous People)	Indigenous People; specific tribe	The Oxford English dictionary defines Indian in two ways: 1) a native or inhabitant of India;* and 2) a member of any of the indigenous peoples of North, Central, and South America, especially those of North America.	* Acceptable use for Indian in this instance
Invalid (noun: in·va·lid)	Avoid in all instances*; use people first language and their diagnosis, ie, "person with X"	The Oxford English dictionary defines an invalid as "a person made weak or disabled by illness or injury." It is probably the oldest term for someone living with physical conditions that are considered seriously limiting.	* Except when used in a direct quote

Lame	“difficulty walking”	Commonly used to describe difficulty walking as the result of an injury to the leg.	
LGBT (when talking with those who are unfamiliar with the issues or are not supportive of the issues)	Gay and transgender; lesbian, gay, bisexual, and transgender*	Per “The Ally’s Guide to Terminology”: Reference sexual orientation and gender identity when talking about issues pertaining to both. (See Transgender for more information.) The abbreviation “LGBT” can be confusing and alienating for those who are unfamiliar with the issues or not yet supportive—though it is essential when talking to LGBT and strongly supportive audiences. Use the term that allows your audience to stay focused on the message without creating confusion about your intended meaning.	* If needed for clarity
Man hours	Work hours		
Man hours/man weeks	Work hours/work weeks		
Man week	Work week		
Manned	Crewed; piloted; human		
Manned aircraft	Crewed aircraft; piloted aircraft; human aircraft	Preferred gender inclusive terms by NASA	
Manpower	Workforce; laborforce		
Master/Slave	leader/follower; parent/child; primary/secondary; main/secondary*		* Preferred term from IEEE Thesaurus
Mentally ill/ emotionally disturbed	Person with a psychiatric disability		
Minorities	People of color*; underrepresented groups†; marginalized	The use of “minority” may be viewed pejoratively because it is usually equated with being less than, oppressed, or deficient in comparison with the majority (i.e., White people).	* POC (people of color). † When possible, use the specific name of the group or groups to which you are referring.
Normal People	Person without X; the majority		

Oriental (when referencing race)	Asian*; Asian American† Avoid in all instances when referring to race	“Orientals” is considered pejorative; be more specific by providing nation and region of origin (Japanese, Chinese, Vietnamese, etc.).	* For people from Asia; † People of Asian descent in North America
Retarded/Slow learner	Learning disability* Avoid in all instances	According to the University of Kansas Research & Training Center on Independent Living: “...describes a neurologically based condition that may manifest itself as difficulty learning and using skills in reading (called dyslexia), writing (dysgraphia), mathematics (dyscalculia) and other cognitive processes due to differences in how the brain processes information. Individuals with learning disabilities have average or above average intelligence, and the term does not include a learning problem that is primarily the result of another cause, such as intellectual disabilities or lack of educational opportunity.”	* Only when the condition has been medically diagnosed
Schizophrenic/Schizo	Person with schizophrenia		
Sex change (operation)	Transition	Per hrc.org: The process by which some people strive to more closely align their internal knowledge of gender with its outward appearance. Some people socially transition, whereby they might begin dressing, using names and pronouns, and/or be socially recognized as another gender. Others undergo physical transitions in which they modify their bodies through medical interventions.	Transition is the accurate term that does not fixate on surgeries, which many transgender people do not or cannot undergo. Terms like “pre-op” or “postop” unnecessarily fixate on a person’s anatomy and should be avoided.
Sexual identity/ transgender identity	Gender identity/gender expression	Gender identity is one’s internal sense of gender. Gender expression is how a person outwardly expresses their gender. (Terms are not interchangeable.)	Many transgender people identify as male or female and not simply transgender. Pronouns express this identity: He/him; She/her; They/them.
Sexual Preference/Gay lifestyle/homosexual lifestyle/same-sex attractions/sexual identity	sexual orientation	“Sexual preference” is used by anti-gay activists to suggest that being gay is a choice; therefore, being gay can be changed. Using “lifestyle” insinuates much the same and stigmatizes gay people suggesting their lives should be viewed strictly as sexual.	

Single/double-blind	Single/double-anonymous		Preferred term from IEEE Thesaurus
Transgendered/ a transgender (n.)/ transgenders (n.)/ transvestite/tranny	Transgender	Transgender is an adjective, not a noun. “Trans” as shorthand is often used within the LGBTQ+ community, but not generally understood by general audiences.	Always use a transgender person’s chosen name. Also, a person who identifies as a certain gender should be referred to using pronouns consistent with that gender (he/him, she/her, they/them).
Unmanned	autonomous; uncrewed; remote		
Unmanned aerial vehicles	Autonomous aerial vehicles (AAV)*; remote aerial vehicles (RAV); uncrewed aerial vehicles (UAV)		* Preferred term from IEEE Thesaurus
Unmanned automobiles or cars	Autonomous automobiles; remote automobiles		Preferred term from IEEE Thesaurus
Unmanned underwater vehicles	Autonomous underwater vehicles (AUV); remote underwater vehicles (RUV)		Preferred term from IEEE Thesaurus
Unmanned vehicles	Autonomous vehicles; remote vehicles		Preferred term from IEEE Thesaurus
Unsuccessful suicide	Attempted suicide		
Wheelchair-bound	Wheelchair user/Uses a wheelchair		
White box	Glass box		Preferred term from IEEE Thesaurus
Whitelist; white list	Access list	A whitelist (or white list) is a list or register of entities that, for one reason or another, are being provided a particular privilege, service, mobility, access or recognition. This can be a controversial view for some.	Preferred term from IEEE Thesaurus